Production of 80 tonnes/day of Maleic Anhydride from partial oxidation of n-Butane

By

Hassan Irtaza Kareem (FA18-CHE-005)

Ahmed Mustafa (FA18-CHE-021)

Farasat Ahmed (FA18-CHE-033)

Hafiz Anees ur Rehman (FA18-CHE-067)

Yahya Riaz (FA18-CHE-102)

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A Chemical Engineering Plant Design Project Report Presented to

COMSATS University Islamabad, Lahore Campus

In the partial fulfilment

of the requirement for the Degree of

Bachelor of Science in Chemical Engineering

By

Hassan Irtaza Kareem (FA18-CHE-005)

Ahmed Mustafa (FA118-CHE-021)

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Name of Students	Registration Number
Hassan Irtaza Kareem	FA18-CHE-005
Ahmed Mustafa	FA18-CHE-021
Farasat Ahmed	FA18-CHE-033
Hafiz Anees ur Rehman	FA18-CHE-067
Yahya Riaz	FA18-CHE-102

Supervisor:

Dr. Abdul Razzaq

(Associate Professor in chemical engineering department

Author's Declaration

We hereby declare that this project report "80 MTPD Production of Maleic Anhydride through the partial oxidation of n-Butane using Vanadium Phosphorous Oxide Catalyst" is our own work on the project and the references are cited inside. This report has not submitted been previously by us for any other degree from COMSATS University Islamabad, Lahore Campus, or any other institution for any purpose.

Signature: ……..

Name: Hassan Irtaza Kareem

Dated: ……...

Signature: …….

Name: Ahmed Mustafa

Dated: ……...

Signature: ……..

Name: Farasat Ahmed

Dated: ……...

Signature: ……..

Name: Hafiz Anees ur Rehman

Dated: ……...

Signature: ……..

Name: Yahya Riaz

Dated: ……...

Supervisor's Declaration

I hereby declare that I have read this thesis, and, in my opinion, this thesis is sufficient in terms of scope and quality for the award of Degree of Bachelor of Science in Chemical Engineering.

Signature :…….

Name of Supervisor Dr. Abdul Razzaq

Date :…….

.

DEDICATION

I dedicate my work to my beloved parents, and my whole family for their overall support to me throughout my degree and my entire life.

ACKNOWLEDGEMENTS

I am highly thankful to my supervisor, Dr. Abdul Razzaq, who tried his best in helping us to complete this project in the best way. His overall cooperation guidance to me and my group made our work much easier. I am also thankful to my concerned resource persons for their guidance in many tasks of this project. I hope our work would be a better helping source for the next coming undergraduates.

ABSTRACT

Production of 80 tonnes/day of Maleic Anhydride from partial oxidation of n-Butane

Maleic Anhydride (MAN), an organic compound (Formula: $C_4H_2O_3$), has vast applications in the field of textiles and fibers, preparation of unsaturated polyester resins, in coatings, adhesives and printing, and much more. It was produced by the oxidation reaction of Benzene, but due to the hazardous nature of the production process, the method was shifted to oxidation of n-Butane in the presence of Vanadium Phosphorous Oxide (VPO) catalyst. This method provided with many benefits i.e., raw material is cheaper, more productivity, environmentally friendly raw material, higher selectivity to the desired reaction, gives more yield to the product, etc. The equipment involved in the manufacture of MAN from n-Butane are Tubular Packed Reactor, Flash Vessel, absorber, Distillation Column and Heat Exchangers. The Reaction conditions of n-Butane wit oxygen in presence of VPO, involves 420 $^{\circ}$C and 2.7 Bar. The desired product is separated by various means of separation processes and a purity of 99.8% MAN is achieved through the process.

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LIST OF ABBREVIATIONS

MAN	Maleic Anhydride ($C_4H_2O_3$)
VPO	Vanadium Phosphorous Oxide
CO_2	Carbon Dioxide
N_2	Nitrogen
HAZID	Hazard Identification
HAZOP	Hazard Operability
UPR	Unsaturated Polyester Resin

Chapter 1

Introduction to the Project

1.1. Project Description

This project is about the Production of Maleic Anhydride by the Oxidation of n-Butane, by using the Vanadium Phosphorous Oxide (VPO) catalyst. An old method (oxidation of Benzene" proved to be environment non-friendly and an uneconomical method to produce maleic Anhydride, when another method "Oxidation of n-Butane" to produce Maleic Anhydride, was proven to be much more economical and environment friendly as compared to that of with Benzene. Here are some tasks which we must do in our Final Year Project.

1.2. Our Tasks during the Project

The aim of this project is to design a production plant capable of manufacturing 80 tons/day maleic anhydride from n-butane. It all involves the following steps:

- Process Flow Diagram
- Energy Balance and Mass Balance on equipment
- Designing of equipment (Absorption tower, packed bed reactor, Reflux vessel, distillation column, Heat exchanger, and Flash vessel).
- Instrumentation and process control
- Cost Analysis

1.3. Introduction

Maleic anhydride is an organic compound. Its formula is $C_4H_2O_3$. It has several types of applications which requires several functionalities and properties. MAN is also known as such as 2, 5-furandione, dihydro-2, 5-dioxofuran, toxilic anhydride, and butene dioic anhydride. Structural formula of maleic anhydride.

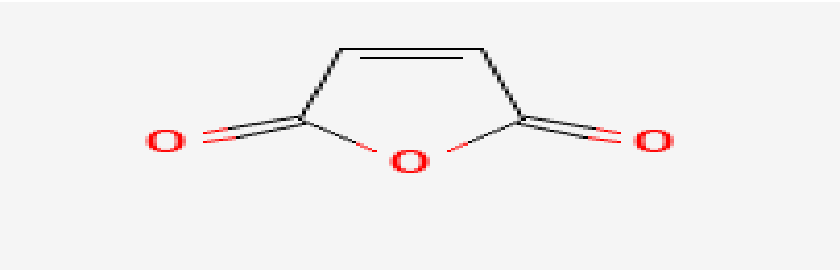


Figure 1.1: Structure of Maleic Anhydride

1.4. Literature Review

It is the acid anhydride of the maleic acid. It was first time synthesized from acid by R. Kempf in 1908. The commercial preparation of maleic anhydride was started in 1930 through the oxidation of benzene. The used of benzene to produce maleic anhydride was dominant until 1980s. There are several methods to produce maleic anhydride but mainly it was produced by two different methods like from butane and benzene. There was also small amount of maleic anhydride being produced as a by-product in the production of phthalic anhydride, when phthalic anhydride was oxidized then we got maleic anhydride.

The usage of benzene to produced maleic anhydride was replaced by n-butane in 1974 as benzene more toxic and effect according to the economical aspect. The first-time maleic anhydride was commercially produced from n-butane at Monsanto's J. F. Queeny plant in the year 1974. The use of benzene was identified as a case of hazards in US. After 1980s, the United States industries that was producing the maleic anhydride underwent a conversion of the feedstock from benzene to butane. During the early years when benzene was replaced by butane as a feedstock butane had limitation of not having a butane-based catalyst which made the production capacity ineffective. A three-year hard work of Monsanto, Halcon and Denka provided an efficient approach to butane to Maleic anhydride Production facility.

It was known as the largest production plant with the increase of production from of 59,000 TPY to 105,000 TPY in 1999. At the mid of the 1980s almost all maleic anhydride was produced by using butane as a feedstock in United State due to increased regulatory pressure, catalyst technology progress and Cost effectiveness of the butane-based process. Meanwhile, European industries that were producing maleic anhydride also shifted benzene to butane-based Production and then constructing the new production facilities by CONDEA Huntsman, Pantochim and Lonza.

In Pakistan NIMIR Industrial Chemicals Limited is the largest producer of maleic anhydride. In 1993 the capacity of fixed bed (butane) was 369ktons/year. But maleic anhydride produced by using fixed bed (benzene) was 325ktons/year. In 2000 the production capacity of fixed bed (butane) was 704ktons/year that was 51% of all type of method producing maleic anhydride through different process and raw materials. In

2018 world production capacity was 3.2 million metric tons. Now a days the demand of maleic anhydride is 3.4 million metric tons.

1.4.1. Physical Properties

This chemical causes a strong irritation to skin, eyes, and mucous membranes of the upper respiratory system. It has irritating and choking odor the compound is soluble in benzene, toluene, and another organic compound from greatest to lowest. It can be shipped both as a solid and in molten state

1.4.2. Chemical Properties

Because of the highly reactiveness due to unsaturated MAN reacts with water to form Fumaric acid and maleic acid at approximately 60 ^{0}C and fumaric acid at 100 ^{0}C. Fumaric acid undergoes crystallization and is not readily soluble in water. Its heat treatment and crystal are mainly used to increase formation fumaric acid by maleic acid isomerization. By fixing aqueous maleic anhydride acid concentration to 40% and the temperature to less than 85 ^{0}C, fumaric acid will not made in less than 0.1% mass of the reaction sample. Maleic acid also thermally losses water from an aqueous solution or by azeotropic distillation and forms maleic anhydride. Maleic anhydride is an organic compound with the chemical formula $C_4H_2O_3$ by having two main carbonyl groups and one double bond. Maleic anhydride is also favorable for photopolymerization and co polymerization making polymers which are known as styrene- maleic anhydride or acyclic acid maleic anhydride. (Doble 2007)

1.4.3. Applications of Maleic Anhydride

Maleic anhydride has a vast range of applications on industrial scale as well as on commercial scale which includes:

- **Used for grafting in polymers:**

Heating the solution of maleic anhydride along-with the certain polymer, generates a macroradical of certain polymer due to the reaction of peroxide with that polymer chain and reaction of polymer-maleic anhydride radical with the polymer chain. These macro-radicals generated may set-off the polymerization of MAN, else join the developing chains of poly-maleic anhydride. Hence, the polymer-grafted maleic anhydride is prepared.

- **Use in Textiles & Fibers:**

Maleic anhydride has an extensive use in textiles and fibers industry. Examples include, increasing the hygroscopicity and good fast washing of cotton fabrics i.e., cotton knits, etc. by treating with a mixture of MAN and polyoxyalkylene copolymer. Also used in the Preparation of Hygroscopic flat yarns used with electric cables. In manufacturing of water-retention sheets used as agricultural goods. Used in the preparation of medical instruments by introducing the extruded polypropylene between two extruded maleic anhydride-grafted liner polyethylene.

- **Use as a coupling agent:**

Maleic Anhydride also helps in blending the two immiscible polymers together to form a blend. Maleic anhydride is highly reactive with the Poly (Lactic Acid) radicals formed by an initiator. An initiator only introduces the free radicals of Poly (Lactic Acid) to react with the maleic anhydride.

- **Introducing into Natural Rubber for improved performance**:

Introduction of maleic anhydride into the natural rubber molecular chain is known as Malleating or Melanization. The rubber resulting from such reaction is known as Malleated rubber or Melanized rubber. Such reactions are carried by two types of reactions i.e., radical reaction and electro cyclic reactions, also called as “ene reaction”. Vulcanized malleated rubbers have high solvent resistance, flex cracking resistance and ageing resistance as compared to non-malleated vulcanized rubbers.

- **Preparation of Unsaturated Polyester Resins:**

Maleic Anhydride has a major use in producing Unsaturated Polyester resins (UPR) which is then used in the pleasure boats, bathroom fixtures, pipes, and tanks. UPR is also used in manufacturing of high-performance parts used in marine and transportation industry i.e., fenders, closure & body panels, hulls, and decks of boats, etc. UPR has a dominant consumption of maleic anhydride due to the low cost, high-performance characteristics, and ease of fabrication of UPR in its vast applications.

- **Use in Diels Alder reaction:**

Maleic anhydride has its applications also in Diels Alder reaction due to its property of being a good dienophile (a reactant in the Diels Alder Reaction), because the tendency of the carbonyl groups to withdraw the electrons makes the two carbons in alkene, to be electron poor, making a good target for the attack of pi electrons in the diene. Diels Alder Reaction has major pharmaceuticals applications and used in biomedical

engineering and also in the synthesis of polymers & dendrimers for drug delivery, functionalized surfaces and preparations of hydrogels for drug delivery.

- **Use in Coatings, Adhesives and Printing:**

In coatings, Maleic Anhydride has vast applications which includes in automobiles, architectural, anti-microbial and anti-adhesives coatings. Coatings contain many key-components like binders, pigments, dispersants, and biocides. Each component has its own specific functionality.

In composite materials, there are two at least two different components. These materials can be made of adhesive resins, fibers, and bi-phase resin systems. The applications of composites range from buildings, structures, transportations, electronics and electrical to furniture, appliances, and sports equipment. They are preferred due to their good stiffness, strength, flexural characteristics, chemical resistance, electrical insulation properties and anti-corrosive nature.

Besides coatings and composite materials, maleic Anhydride also has a scope in printing applications such as its reactivity or cure speed, compositional versatility, providing color and its stability, etc are some of printing applications of Maleic Anhydride. (Hood 2016)

- **Use in food packaging materials:**

Maleic Anhydride also has its applications in manufacturing the food packaging materials such as used in manufacture of recyclable bottles (for juice bottles, milk bottle, etc. (Ansell 2015)

1.4.4. Current Market Demand

The Global Maleic Anhydride market was valued at USD 2.8 billion in 2019 and is now estimated to reach USD 3.5 billion by 2026, expanding at a Compound Annual Growth Rate (CAGR) of around 4.5% during the forecast period i.e., 2020-2026. The rising automotive industry, coatings industry, increasing wind energy and construction industries are expected to contribute to the growth of Maleic Anhydride market around the globe during the forecast period. To produce unsaturated polyester resin, maleic anhydride is an essential raw material. Unsaturated polyester resins are used as a substitute in furniture, fixtures, and construction industry due to its light weight and mechanical strength. Maleic Anhydride is also used in tank manufacturing, marine vessels, automotive parts, and coatings production. Therefore, rise in these industries will also cause a growth in the Maleic Anhydride market. (Doble 2007)

In Pakistan, rise in these industries also demands a growth in the production of Maleic Anhydride. Presently, only Nimir Chemicals Pakistan Limited is the sole producer of Maleic Anhydride in Pakistan. Therefore, due to increasing demand, there is a need of increased Maleic Anhydride production. (Maleic-Anhydride Market Research Report n.d.)

1.4.5. Scope of Maleic Anhydride in Pakistan

In Pakistan, increasing automotive industry, construction industry, pharmaceutical industry, and coatings industry contribute to the growth of maleic anhydride market in Pakistan. The dominant application of maleic anhydride is the production of unsaturated polyester resins which are further used to manufacture composite materials mainly used in the construction industry and automotive industry.

In Pakistan, different industries are using Maleic Anhydride to produce polyester resins. Major consumers of Maleic Anhydride in Pakistan are:

- Descon Engineering Limited are the utilizers of Maleic Anhydride to produce polyester resins.
- Dawood Chemical Industries utilize it in the manufacture of binders and glues, used in manufacture of Plywood. They also produce Polyester resins from MAN.
- Karachi Al-Khair is also producing Polyester Resin from Maleic Anhydride.
- Only Producer of Maleic anhydride in Pakistan country is Nimir Chemicals limited and still utilizing the Benzene based Oxidation Technology which is old and is now replaced with the oxidation of n-Butane by many industries. (Bhatti n.d.) As the production from n-Butane has many environmental and economic benefits as compared to production from Benzene which are listed as follows:
- From an environmental point of view, the process produces about 1/3 of gaseous products without benzene and benzene derivatives and less CO and CO_2; it uses only 10% of process water and, moreover, produces less liquid waste (1.6 m^3/ton of maleic anhydride Vs 3.6 m^3/ton of maleic anhydride referred to benzene process).
- From an economic point of view, maleic anhydride production cost is lower (60%) compared to that of the benzene process. The maleic anhydride grade is 99.9999%, as specifically required in some applications. (Bhatti n.d.)

Due to easy availability, lower production cost, and less health hazards of n-Butane-based maleic anhydride, it is preferable both economically and environmentally.

Therefore, due to increased demand of maleic anhydride and benefits of n-Butane based maleic anhydride, Maleic Anhydride production from n-Butane is more demanding in Pakistan.

1.5. Process Selection among various processes

Maleic Anhydride is produced by the oxidation reaction of benzene by passing over a vanadium Pentoxide (V_2O_5) catalyst or by oxidation reaction of n-butane by passing over Vanadium Phosphorous Oxide (VPO) catalyst. The details for the respective processes are given below:

1. Maleic Anhydride production by using Benzene.
2. Maleic Anhydride production by using n-butane.

The description of the processes is described below:

1.5.1. Maleic Anhydride production by using Benzene

Although, benzene is an expensive, highly toxic and an environmental hazardous chemical, but it has been used largely to produce Maleic Anhydride before 1980. Basically, Maleic Anhydride was produced by the oxidation of benzene, passing it over V_2O_5 catalyst (which was modified with MoO_3 and H_3PO_4). It was a vapor-Phase oxidation reaction carried out in a packed bed reactor. The temperature and pressure conditions maintained for the production process were (400-450)^{0}C and 2 to 5 bars respectively.

1.5.2. Maleic Anhydride production by using n-Butane

Butane is a cheaper, non-toxic, environmentally friendly hydrocarbon. Only it is now being used to produce Maleic Anhydride due to its economic, process and environmentally friendly nature. Basically, Maleic Anhydride is produced by the oxidation of butane passing it through the Vanadium Phosphorous catalyst (VPO i.e., a heterogenous catalyst). The by-products formed by the reaction include CO and CO_2. It is Vapor-phase reaction carried out in Packed Bed Reactor at a temperature of (350 to 500)^{0}C and pressure of 2.75 bars. (Modak 1998). The comparison for both the processes is given in Table 1.*1*, while the Merits and Demerits of the production processes for Maleic Anhydride are given in Table 1.*2* and Table 1.*3* respectively.

Table 1.1: Comparison table of both production processes

Sr. #	Process Parameters	Oxidation of Benzene.	Oxidation of n-Butane.
1.	**Raw Materials**	Benzene, Oxygen.	Butane, Oxygen.
2.	**Economics**	Expensive.	Cheap.
3.	**Toxicity**	Highly reactive and toxic due to the presence of double bonds.	Non-toxic and environment friendly.
4.	**Heat evolved**	Less Exothermic.	Highly Exothermic.
5.	**Raw material conversion achieved**	85-95%.	85%.
6.	**Yield**	Less due to high consumption.	More due to less consumption.
7.	**Temperature Required**	400-450^{0}C.	350-500^{0}C.
8.	**Pressure Required**	2-5 bars.	2.75 bars.

Table 1.2: Merits of both processes

Sr. #	Oxidation of n-Butane	Oxidation of Benzene.
1.	The raw material is cheaper.	It evolves less amount of heat.
2.	1 Tonne n-butane produces 1 Tonne Maleic Anhydride.	85-95 % Benzene conversion is achieved.
3.	Environment friendly Raw Material.	Somehow low temperature range (400-450^{0}C) is required.
4.	Less amount of Oxygen Required for oxidation.	
5.	Selectivity to Maleic Anhydride is 70%.	
6.	Gives 30 % more yield of Maleic Anhydride with n-Butane.	

Table 1.3: Demerits of Both Production Processes

Sr. #	Oxidation of n-Butane.	Oxidation of Benzene.
1.	It is a highly exothermic reaction.	It is 1.5 times more expensive than that of n-butane.
2.	85% of raw material conversion is achieved.	1.11 Tonnes of Benzene produces 1 Tonne of Maleic Anhydride.
3.	Somehow high Temperature range (350-500^{0}C) is required.	More Oxygen consumption.
4.		Highly toxic and Environmental Hazardous
5.		Selectivity to Maleic Anhydride obtained is 60-65%
6.		Gives 30% lesser yield of Maleic anhydride with Benzene.

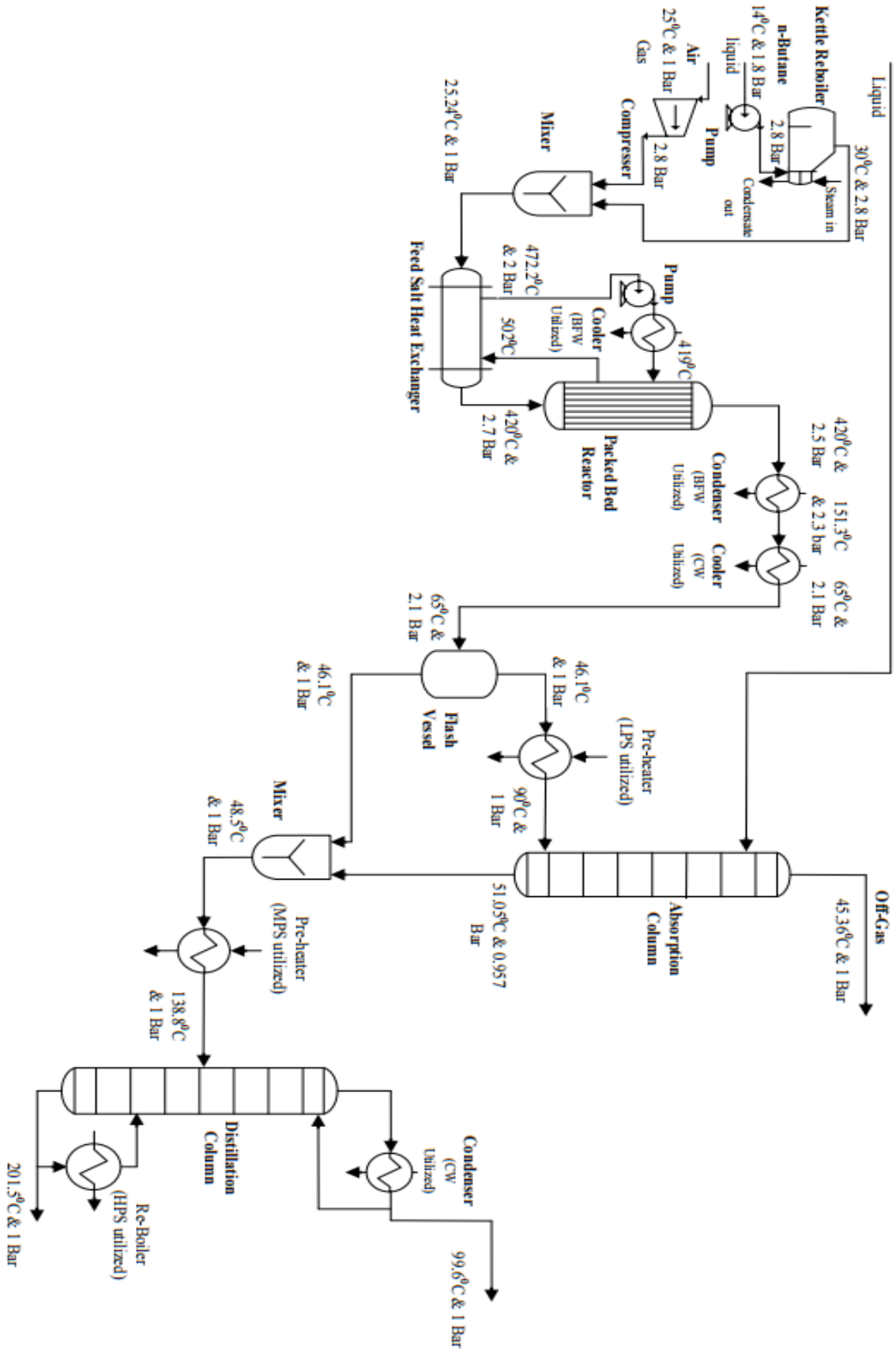
1.5.3. Process Selected

The process which we have selected to produce Maleic Anhydride is **Oxidation of n-Butane**. As this process has the following characteristics:

- The Raw material used in this process i.e., n-Butane, is a cheaper hydrocarbon and provides higher yield.
- Less quantity of raw material (n-Butane) is used, as compared to that of benzene.
- N-butane is environment friendly raw material as compared to Benzene (Toxic).
- Selectivity obtained by using n-Butane is higher than that of Benzene.
- All 4 carbon-atoms of n-butane are utilized in reaction, while all 6 carbon-atoms of Benzene are not utilized.

1.6. Process Flow Diagram and its Description

Process Flow Diagram:



Description of Process Flow Diagram:

The process begins with fresh butane which is pressurized at from 1.8 to 2.8 bar from a pump and then vaporized to 30^0C at 2.8 bar and mixed with the pressurized air at 2.8 bar at 25^0C and moved towards FSHE.

In this section the mixture is preheated in FSHE (Feed-Salt-Heat-Exchanger) to 420^0C before entering the tubular reactor and to be reminded that the hot fluid in the exchanger is a molten salt solution of sodium nitrate and nitrite. Which has a multipurpose to preheat the mixture of air and butane and to reduce the temperature of the exothermic reaction of butane and air with Vanadium phosphorus oxide (VPO) as a catalyst.

The outcome of the reactor is MAN and many side products including (N2, H2O, CO_2, C4 and $O_{2)}$ at 420^0C and 2.6 bar and then the mixture is forwarded to a series of heat exchangers (condenser followed by a cooler) where it temperature drops to 65^0C and then the steam is opened in a flash separator and pressure drops to 1 atm. (Note: the top stream of the flash vessel is pre-heated to 90^0C before entering the absorber). Where the vapor is moved to the absorber and the bottom is forwarded to the distillation column.

In the absorber section water sprinkled in the column at a rate of 2438 kg/hour and 30^0C. MAN and water comes out the bottom where it is mixed with the bottom product of the flash separator. The Mass fraction of water is (0.619 H_2O and 0.3805 MAN) at 1 bar and a flow rate of 8759.47 kg/hour it should be noted that the flow rate is recorded after it has been passed from the pump.

The top product of the absorber is divided, and the unwanted off-gas is sent to flare and the in the bottom of the absorber the product is pre-heated and send to the distillation column at 138.8°C.

Now, the bottom product of the absorption and the flash vessel is sent to the binary distillation column where the water and the maleic anhydride is separated into two products.

- Top product
- Bottom product

Top product:

The product consists of ≈ 0.002 mass faction of water and remaining is MAN (maleic anhydride).

Bottom product:

The bottom product consists of 0.998 % MAN and 0.002 water.

1.7. Hazards Identification (HAZID)

This chapter includes the Hazards associated with the production of Maleic Anhydride, their sources, effects on the production process and their control measures to be taken, briefly described below:

1.7.1. Hazards Associated, their Sources, Effects & Control Measures

Butane based production of MAN is based upon the requirement and presence of high temperature and pressure. Also, n-Butane which has a flammability, is utilized in this (MAN) production plant. Hence this process becomes much more hazardous to the environment and the working people around and the equipment involved. The relevant hazards involved in the production of MAN are described below according to its common source, and their aftereffects and their controlling measures are described in the table below: (Maleic Anhydride 2016)

Table 1.4: Hazards, their Sources, Effects and Control Measures

Source	**Potential Hazards**	**Effects**	**Control Measures**
Chemical Tank, Storage & Transportation	Dangerous chemical reaction, overpressure, temperature above flash point, corrosion of storage tank, release of	• Fire • Explosion • Air pollution • Land pollution	• Replace hazardous chemical with a less hazardous alternative. • Isolation of incompatible chemicals • Installation of pressure safety

	toxic fume, chemical spillage		valve • Filtration of toxic fume • Warning signage/ label for chemicals • Store chemical in well-ventilated area
Reactor	Overpressure, overheat, sudden temperature rise	• Explosion	• Utilize overpressure relief protection • Implement control system to prevent overheat and overpressure condition • Installation of pressure safety valve
Distillation Column	High pressure	• Flooding	• Sequence the distillation process for a minimum flow of non-key components • Utilize overpressure relief protection

Heat Transfer	Overpressure, fouling, overflow of cooling or heating medium	• Tube rupture • Leaking • Contamination of liquid • Thermal shock	• Utilize overpressure relief protection • Substitution to a less hazardous cooling or heating medium • Use control valve to prevent thermal shock
Utilities (Electricity, gas, water, etc.)	Short circuit, ignition	• Fire • Explosion	• Ensure proper connections and maintenance for electrical components • Keep flammable material away from ignition source
Layout	Unsafe layout (poor arrangement, limited space to perform safe work)	• Create less conducive, accident-prone workplace	• Ensure safe and practical design of process plant, considering safe operational sequence and future expansion • Separate process and non-process area. • Ensure proper ventilation
Location	Natural disaster,	• High risk of disaster	• Selection of site with minimal

	approval from local community	• Pollution affecting community (noise, air, etc.)	risk of natural disaster. • Keep safe distance between the plant site and the local community
Environment	Untreated wastewater	• Water pollution	• Develop a treatment system for wastewater

Chapter 2

Material Balance

2.1. Primary Calculations

Basis: 80MT/day of Maleic anhydride.

80 MT	1000kg	1 Day
Day	1MT	24 hours

Maleic Anhydride to be Produced = 3333.333 kg/h.

Balance Chemical equation:

$$C_4H_{10} + 3.5O_2 \text{ -----> } C_4H_2O_3 + 4H_2O$$

Quantity of Maleic Anhydride to be produced	= 80 tonnes/ day
Molar Mass of Oxygen	= 32 kg/kmol
Molar Mass of CO_2	= 44 kg/kmol
Molar Mass of Water	= 18 kg/kmol
Molar Mass of n-Butane	= 58 kg/kmol
Molar Mass of Maleic Anhydride	= 98.06 kg/kmol
Quantity of Maleic Anhydride to be produced	= 3333.333 kg/h
Quantity of Maleic Anhydride to be produced	= 33.99279 kmol/h

The Butane in an oxidation reaction has 85% conversion at its maximum yield of product.

2.2. Overall Mass Balance

From Balanced Chemical Equation:

Selectivity for this reaction = 85%

Total Conversion of butane = 85%

Conversion of butane for the above reaction is = 72.2500%

Butane required to produce 3333.33 kg/hr MAN:

1971.582kg/h OR 33.993 kmol/h of n-Butane

Actual Amount of n-Butane required:

2728.8333kg/h OR 47.0488 kmol/h of n-Butane

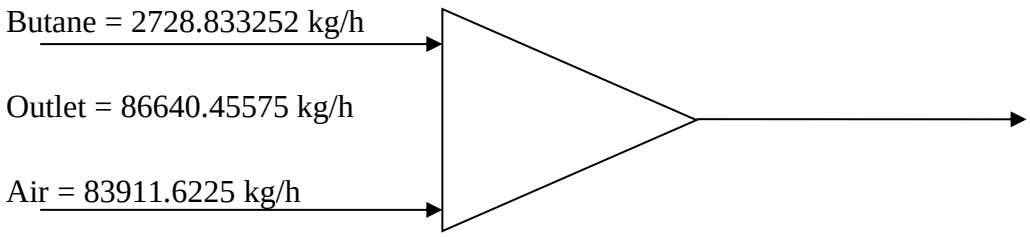
Table 2.1: Amount of Air required in Feed

Lower Flammable limit of n-Butane in Air	= 1.8 mol%
Assumed Composition of n-Butane in Feed Air	= 1.6 mol%
Molar Mass of Air	= 29 kg/kmol
100 kmol/h of Air contains	= 1.6 kmol/h OR 92.8 kg/h of n-butane
100 kmol/h of Air contains	= 98.4 kmol/h OR 2853.6 kg/h of Air
92.8 kg/h of n-Butane requires	= 2853.6 kg/h of Air
2728.833 kg/h of n-Butane requires	= 83911.623kg/h OR 2893.51 kmol/h Air

2.3. Mass Balance around Mixer 1

Table 2.2: Mass Balance around Mixer 1

Inlets	**Components**		**Flow rates**		**Compositions**	
			kmol/h	**kg/h**	Molar	Mass
Stream 1	Butane	n-Butane	47.04884917	2728.833252	1	1
		Total	47.04884917	2728.833252		
Stream 2	Air	Oxygen	607.6358871	17621.44073	0.21	0.21
		Nitrogen	2256.933295	65451.06555	0.78	0.78
		CO_2	28.93504224	839.116225	0.01	0.01
		Total	2893.504224	83911.6225	1	1
Outlet Streams	Butane		47.04884917	2728.833252	0.016	0.031496063
	Nitrogen		2256.933295	65451.06555	0.76752	0.755433071
	Oxygen		607.6358871	17621.44073	0.20664	0.203385827
	CO_2		28.93504224	839.116225	0.00984	0.009685039
	Total		2940.553073	86640.45575	1	1



2.4. Mass Balance around Reactor

Reaction 1:

$$\mathbf{C_4H_{10} + 3.5O_2 \text{ -----> } C_4H_2O_3 + 4H_2O}$$

Conversion of Butane in this reaction = 72.25%

From Balanced Chemical Equation:

Butane Converted to MAN = 1971.582025 kg/h

Maleic Anhydride produced = 3333.333333 kg/h

Water Produced = 2447.481134 kg/h

Oxygen Used = 3807.192875 kg/h

Reaction 2:

$$\mathbf{C_4H_{10} + 6.5O_2 \text{ -----> } 4CO_2 + 5H_2O}$$

Selectivity for this reaction = 15%

Conversion of Butane in this reaction = 12.7500%

From balanced Chemical Equation:

Butane Converted to Carbon Dioxide = 347.9262396 kg/h

Carbon Dioxide produced = 1055.776175 kg/h

Water Produced = 539.8855443 kg/h

Oxygen Used = 1247.73548 kg/h

Mass Balance Equation for a Reactor:

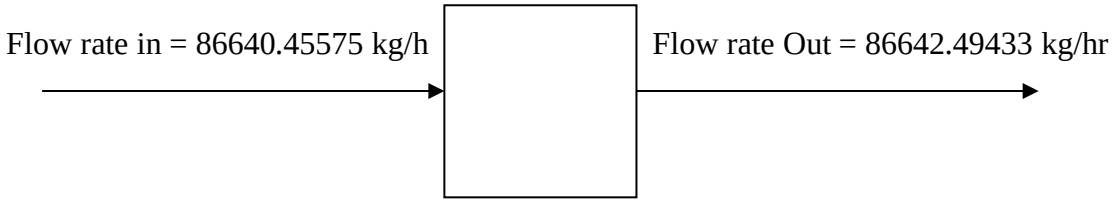
Accumulation = Flow rate In- Flow rate Out + Generation - Consumption

Accumulation = 0.000987808

Hence, Accumulation is approximately = 0

Table 2.3: Mass balance around Reactor

Components	**Flow rates In (kg/h)**	**Flow rates out (kg/h)**	**Flow rates gen. (kg/h)**	**Flow rates cons. (kg/h)**
CO_2 =	839.116225	1894.8924	1055.776175	0
O_2 =	17621.44073	12566.51237	0	5054.928355
N_2 =	65451.06555	65451.06555	0	0
H_2O =	0	2987.366678	2987.366678	0
C_4H_{10}=	2728.833252	409.324	0	2319.508264
MAN =	0	3333.333333	3333.333333	0
Total =	86640.45575	86642.49433	7376.476187	7374.4365



2.5. Mass Balance on the Salt Cooling System (Salt stream)

Table 2.4: Mass balance on Salt Cooling System

Components	**Molar Mass** kg/kmol	**Molar Composition**	**Molar Flow rates** kmol/h	**Mass Flow rates** kg/h
$NaNO_2$	68.995	0.413	2065	142474.675
$NaNO_3$	84.9947	0.587	2935	249459.445
Total =		1	5000	391934.1195

2.6. Material Balance around Flash Vessel

Table 2.5: Mass Balance around Flash Vessel

Component Names	**Component Alias**	**Flow rate (Kg/h)**	**Mass fraction**	**Flash Down (Kg/h)**	**Flash down composition (Kg/h)**	**Flow Top (Kg/h)**	**Flash Top Composition**
Maleic Anhydride	$C_2H_2O_3$	3333.333	0.03847	2935.82	0.92	397.52	0.00476
Water	H_2O	2987.366	0.03447	255.13	0.0799	2732.236	0.03274
Carbon dioxide	CO_2	1894.892	0.0218	0	0	1894.8924	0.0227
Oxygen	O_2	12566.512	0.14503	0	0	12566.512	0.1505
Nitrogen	N_2	65451.065	0.75541	0	0	65451.065	0.7843
n-Butane	C_4H_{10}	409.324	0.004724	0	0	409.324	0.0049
Total		86642.494	1	3190.95	1	83451.55	1

2.7. Material Balance around the Absorber

Table 2.6: Mass Balance around Absorber

Component	**Component**	**Mass Fractions**	**Flow rate In**
Maleic Anhydride	$C_2H_2O_3$	0.00476	397.52
Water	H_2O	0.03274	2732.236
Carbon dioxide	CO_2	0.0227	1894.8924
Oxygen	O_2	0.1505	12566.512
Nitrogen	N_2	0.7843	65451.065
n-Butane	C_4H_6	0.0049	409.324

Component	Absorber outlet Top	Absorber inlet bottom	Absorber Top inlet	Absorber outlet bottom	Absorber Outlet Bottom Mass Fraction
Maleic Anhydride	0	397.52	0	397.52	0.0714
Water	0	2732.236	2438.77	5170.756	0.9286
Carbon dioxide	1894.8924	1894.8924	0	0	0
Oxygen	12566.51237	12566.51237	0	0	0
Nitrogen	65451.06555	65451.06555	0	0	0
n-Butane	409.324	409.324	0	0	0
Total	80321.79432	83451.55	2438.77	5568.276	1

Absorber Efficiency = 100%

Solubility of MAN = 16.3 kg of MAN in 100kg of water

Maleic Anhydride to be absorbed = 397.52 kg/h

Water required = 2438.77 kg/h

2.8. Mass Balance around Mixer 2

Table 2.7: Mass Balance around Mixer 2

Component	Fraction from Absorber	Rate from Absorber	fraction from Flash	Flow rate of flash bottom	Total	Mass fraction
MAN	0.0714	397.52	0.92	2935.82	3333.33	0.38
Water	0.9286	5170.756	0.0799	255.13	5426.139006	0.62
Total	1	5568.276	1	3190.95	8759.479	1

2.9. Mass Balance around Distillation Column

Table 2.8: Mass Balance around Distillation Column

Component	**Mass Flow rate In**	**Mass Fraction**	**Bottom Product**	**Mass fraction**	**Top Product**	**Mass Fraction**
Maleic Anhydride	3333.333	0.62	3326.66	0.998	10.85	0.002
Water	5426.139	0.38	6.6667	0.002	5415.29	0.998
Total	8759.479	1	3333.34	1	5426.146	1

Hence, Flow rate of the Bottom Product of the Distillation Column i.e., 3333.33 kg/h (80 Tonnes/day) is the Flow rate of Maleic Anhydride to be produced with 99.8% Purity.

Chapter 3

`Energy Balance

3.1. Energy Balance around Reactor

3.1.1. For Salt Cooling system

Table 3.1: Energy balance on Salt Cooling system

Components	Cp = Specific heats kJ/kg K		mCp kJ/h K	
	419^0C & 2 bar	**502.2^0C & 2 bar**	**419^0C & 2 bar**	**502.2^0C & 2 bar**
$NaNO_2$	2.174	2.2761	278765.9491	291857.947
$NaNO_3$	1.2682	1.2842	284728.0208	288320.2368
Total =			563493.9699	580178.1838

Reference temperature = 273 K

Inlet Temperature = T1 = 692 K

Outlet temperature = T2 = 775.2 K

Heat In = mCp (T1-Tref) =236103973.4 kJ/h.

Heat Out = mCp (T2-Tref) = 291365483.9 kJ/h.

Heat removed by the salt = Heat In -Heat Out = 55261510.51 kJ/h

3.1.2. For Reactor Inlet @ 420ºC & 2.7 bar

Table 3.2: Heat taken in by Reactor Inlet Feed

Components	Mass flow rates (kg/h)	Cp (kJ/kg K)	mCp (kJ/h K)
		@ 420^0C & 2.7 bar	**@ 420^0C & 2.7 bar**
Butane	2728.8332	3.4525	9421.296623
CO_2	839.116225	1.1293	947.6139529
Nitrogen	65451.06555	1.0989	71924.17593
Oxygen	17621.4407	1.029	18132.46248
Total =	86640.45568		100425.549

Reference temperature = 273 K

Inlet Temperature = T1 = 693 K

Heat taken in by the Reactor feed = mCp (T1-Tref) = 42178730.58 kJ/h.

3.1.3. For Reactor Outlet @ 420^{0}C & 2.6 bar

Table 3.3: Heat Taken out by Reactor Outlet Products

Components	**Mass flowrates (kg/h)**	**Cp (kJ/kg K)**	**mCp (kJ/h K)**
		@ 420^{0}C & 2.6 bar	**@ 420^{0}C & 2.6 bar**
Butane	409.324	3.4525	1413.19111
CO_2	1894.8924	1.1172	2116.973789
Nitrogen	65451.065	1.0965	71767.09277
Oxygen	12566.5123	1.029	12930.94116
Water	2987.366	2.0789	6210.435177
MAN	3333.33	1.7704	5901.327432
Total =	80321.7937		100339.9614

Reference temperature = 273 K

Inlet Temperature = T2 = 693 K

Heat taken in by the Reactor feed = mCp (T2-Tref) = 42142783.8 kJ/h.

3.1.4. Heat generated in the reactor

Heat generated in the Reactor = Heat Out - Heat In + Heat Removed.

Heat generated in the Reactor = 55225563.74 kJ/h

3.2. Energy balance around Flash Vessel

3.2.1. Heat taken in by the Inlet Stream @65^{0}C & 2.1 bar

Table 3.4: Heat Taken into Flash Vessel by the Feed

Components	**Mass Flowrates (kg/h)**	**Cp (kJ/kg K)**	**mCp (kJ/h K)**
		65^{0}C & 2.1 bar	**65^{0}C & 2.1 bar**
MAN	3333.33	1.38	4499.995
CO_2	1894.89	0.9409	1782.90
Oxygen	12566.512	0.9125	11466.9422
Nitrogen	65451.065	1.68	109957.7892
Water	2987.366	4.1872	12508.69892
Butane	409.324	1.9238	787.457
Total	86642.487		141003.7853

Reference temperature = 273 K

Inlet Temperature = T_1 = 338 K

Heat taken in by the Inlet Stream = mCp (T_1-Tref) = 9165246.046 kJ/h.

3.2.2. Heat Taken out by the Top Product of Vessel @46.1^{0}C & 1 bar

Table 3.5: Heat taken out by the gas stream

Components		**Cp kJ/(kg K)**	**mCp kJ/(h K)**
		46.1^{0}C & 1 bar	**46.1^{0}C & 1 bar**
MAN	397.52	1.528	15.28
CO_2	1894.89	0.8708	1650.070212
Oxygen	12566.512	1.052	13219.97062
Nitrogen	65451.065	1.04068	68113.61432
Water	2732.236	4.2117	880.7296455
Butane	409.324	1.782	729.415368
Total	83451.55		84609.08017

Reference temperature = 273 K

Inlet Temperature = T_2 = 319.1 K

Heat taken out by the TOP Outlet Stream = mCp (T_2-Tref) = 3900478.596 kJ/h.

3.2.3. Heat Taken out by the Bottom Product of Vessel @46.1^{0}C & 1 bar

Table 3.6: Heat taken out by liquid stream

Components	**Mass Flowrates (kg/h)**	**Cp kJ/(kg K)**	**mCp kJ/(h K)**
		46.1^{0}C & 1 bar	**46.1^{0}C & 1 bar**
MAN	3323.33	1.528	5078.04824
Water	2778.25	4.2117	11701.15553
	6101.58		16779.20377

Reference temperature = 273 K

Inlet Temperature = T_2 = 319.1 K

Heat taken out by the BOTTOM Outlet Stream = mCp (T_2-Tref) = 773521.2936 kJ/h.

3.2.4. Heat evolved by the separation in Flash Separator

Heat absorbed by separation in the Flash separator = heat In by the Feed stream - Heat Out by Top Stream + Heat Out by Bottom stream.

Heat absorbed by separation in the Flash separator = 3974061.357 kJ/h.

3.3. Energy balance around Distillation Column

3.3.1. For Inlet feed stream @138.8^{0}C & 1 bar

Table 3.7: Heat taken in by the Distillation column Feed

Components	Cp (kJ/kg K)	Flow rate (kg/s)	Temperature (K)	Q (KW)
MAN	1.751	0.9259	411.8	225.053
water	4.2823	1.5072	411.8	895.89
Total =				1120.926

3.3.2. For Distillate Product @99.6^{0}C & 1 bar

Table 3.8: Heat taken out by the Distillate

Components	Cp (kJ/kg K)	Flow rate (kg/s)	Temperature (K)	Q (KW)
MAN	1.467	0.00301	372.6	0.44046
Water	4.216	1.50424	372.6	631.654
Total =				632.0951

3.3.3. For Bottom Product @201.5^{0}C & 1 bar

Table 3.9: Heat taken out by the Bottom Products

Components	Cp (kJ/kg K)	Mass Fraction	Temperature (K)	Q (KW)
MAN	1.909	0.9240	474.5	355.457
Water	4.46	0.001851	474.5	1.6642

3.4. Energy Balance around Absorber

3.4.1. For Water Inlet Stream

Table 3.10: Heat taken in the water stream

Components	Cp (kJ/kg K)	Flow rate (kg/s)	Temperature (K)	Q (KW)
Water	4.197961667	0.6774	303	85.3156

3.4.2. For Absorber Inlet Bottom Feed Stream

Table 3.11: Heat taken in by the gas stream

Components	Cp (kJ/kg K)	Flow rate (kg/s)	Temperature (K)	Q (KW)
MAN	1.64042	0.11042	363	16.3025
water	4.19325	0.75895	363	286.4238
CO_2	0.911	0.52635	363	43.1561
Oxygen	0.932	3.4906	363	292.799
Nitrogen	1.0428	18.18085	363	1706.3092
n-Butane	2.0007	0.113701	363	20.47336
Total		23.1809		20.4733

3.4.3. For Absorber Top Product

Table 3.12: Heat taken out by the Top stream

Components	Cp (kJ/kg K)	Flow rate (kg/s)	Temperature (K)	Q (KW)
Air	1.00729	22.19790842	318.36	1014.2374
n-Butane	1.77891	0.113701111	318.36	9.17469
Total =				1023.4121

3.4.4. For Absorber Bottom Product

Table 3.13: Heat taken out by the bottom stream

Components	Cp (kJ/kg K)	Flow rate (kg/s)	Temperature (K)	Q (KW)
MAN	1.5423411	0.11042	324.05	8.69426
water	4.048388667	1.43639	324.05	296.859
Total =				305.5536

3.5. Energy Balance around Feed Salt Heat Exchanger

3.5.1. For Coolant inlet (Vertical in 1)

Table 3.14: Heat taken in the Coolant inlet

Components	Cp (kJ/kg K)	mCp (kJ/s K)	Q (KW)
	502^0C & 2 bar	**502^0C & 2 bar**	
$NaNO_2$	2.278	90.079	45219.9658
$NaNO_3$	1.284	88.987	44671.839
Total =		179.0673	89891.805

3.5.2. For Coolant outlet (Vertical out 2)

Table 3.15: Heat taken out the Coolant outlet

Components	Cp kJ/kg K	mCp (kJ/s K)	Q (KW)
	472.2^0C & 2 bar	**472.2^0C & 2 bar**	
$NaNO_2$	2.232	88.33429	41711.4557
$NaNO_3$	1.278	88.5581	41817.136
Total =		176.8924	83528.5918

3.5.3. For Hot inlet 3

Table 3.16: Heat taken in by the Hot inlet stream

Components	Cp (kJ/kg K)	flow rate (kg/s)	Temperature (K)	Q (KW)
Air	1.00652	23.3087833	298	586.5189
butane	1.76517	0.54766167	303	29.0014
Total =				615.5203

3.5.4. For Hot outlet 4

Table 3.17: Heat taken out by the Hot outlet Stream

Components	Cp (kJ/kg K)	flow rate (kg/s)	Temperature (K)	Q (KW)
Air	1.06958	23.30878403	693	10470.855
butane	3.20623	0.758009237	693	1020.7478
Total				11491.602

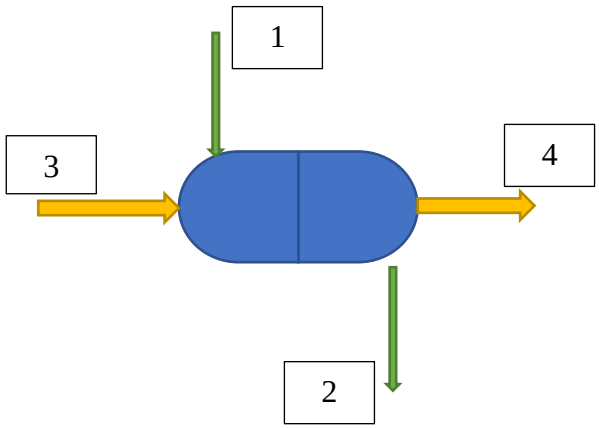


Figure 3.1: Feed salt heat exchanger

3.6. Energy Balance around Compressor (Before FSHE)

Inlet Conditions:

T= 25°C

P= 1 bar

Outlet Conditions:

P= 2.8 bar

T=?

Compressor

Efficiency for the compressor is Ep= 74.5%

For compression ratio= 2

Isentropic Efficiency of Compressor (Richard & Coulson):

Epi= 78%

In Industry, compressor path will be polytrophic, as approximated by the equation:

PV^n =constant

For Polytrophic process, the ratio of inlet temperature and outlet temperature is:

$(T_1/T_2) = (P_1 / P_2)^m$

Where m =(r-1) (r * Ep) at conditions away from critical

Critical conditions for air are:

Pc =37.2 atm= 37.8bar

Tc =-140.52^{0}C

For air (as diatomic gases)

r= 1.4

m= (1.4-1) x (1.4*0.745)

m= 0.4172

Now, for Polytrophic Process:

$(T_1/T_2) = (P_1 / P_2)^m$

Outlet Temperature:

$T_2 = T_1 (P_2/P_1)^m$

$T_2 = 298(2.8/1)^{0.4172}$

$T_2 = 458K = 185°C$

For Centrifugal compressors, the work done will be:

$W = (P_1V_1)\ (n/n-1)\ ((P_2/P_1)^{n-1/n} - 1)$

$W = (RT_1/M)\ (n/n-1)\ ((P_2/P_1)^{n-1/n} - 1)$

n=r

(For isentropic conditions)

For ideal conditions, work done will be:

$W = P_1V_1 \ln (P_2/P_1)$

$m = (z \times r/C_p)\ (1/E_p + x)$

Mean specific heat is to be calculated at mean temperature & mean pressure:

$(T_1 + T_2)/2 = 378K = 105^0C$

$(P_1 + P_2)/2 = 1.9$ bar

$C_p = 29.44$ kJ/kmol K

(Compressibility factor) Z=1

For ideal gases and low pressure:

X=0

Y=1

m= (1*8.314/29.44) (1/0.745 + 0)

m= 0.379

Thus,

n= [1/(y-m (1+x)]

n= 1.61

Thus, for centrifugal compressor the work done will be:

$W = (RT_1/M)\ (n/n-1)\ ((P_2/P_1)^{n-1/n} - 1)$

W= 108 kJ/kg of air

Hence,

Actual Work done = 108/0.745 = 145 kJ/kg of air

W_{ACTUAL} = 145 kJ/kg of air

Chapter 4

Equipment Design

4.1. Feed Salt Heat Exchanger Design

The reactor feed needs to be preheated up to 420^0C (due to reaction conditions) before entering the reactor. The stream from the Mixer contains a mixture of n-butane, nitrogen, oxygen, and carbon dioxide. This stream exchanges the heat with the hot fluid (Molten Salt, mixture of $NaNO_2$ and $NaNO_3$) and then stream containing n-butane enters the Reactor. The Molten Salt is within the jackets of Reactor that gain heat from there and then exchanges it with n-butane-air mixture and again flow back to the reactor. This process continues.

4.1.1. Selection of type of Heat Exchanger

The most common type of Heat Exchanger used in the process industries is Shell & Tube heat exchanger because:

- It is suitable for large heat transfer areas mainly greater than $200ft^2$.
- It gives large surface area in a small volume (less floor space) which helps in the saving of material of construction for large heat transfer areas when compared with Double-Pipe heat exchanger.
- Also, it uses well established fabrication techniques, therefore its maintenance is easy, and it is easier to clean.

To select which type of heat exchanger is suitable for our system, we firstly proceed with the general equation for heat transfer across a surface and find the heat transfer area, if heat transfer area > $200ft^2$ and it meets other specifications, then we will proceed with a Shell & Tube Heat Exchanger design.

The general equation for heat transfer across a surface is

$Q = UA\Delta T_m$ (KERN n.d.)

To find the heat transfer surface area, we proceed as follows:

Parameters for Hot Fluid (Molten Salt) & Cold Fluid (Air-n-Butane mixture):

For Hot Fluid:

Heat Capacity = C_p = 128942 J/ kmol. K

Inlet Temperature = T_1 = 502°C

Outlet Temperature = T_2 = 472.28027°C

Molar Flow Rate = n = 5000 kmol/h.

For Cold Fluid:

Heat Capacity = C_p = 31190 J/kmol. K (the NIST WebBook n.d.)

Inlet Temperature = t_1 = 25.2443° C

Outlet Temperature = t_2 = 420°C

Molar Flow Rate = n = 1556.2 kmol/h.

4.1.2. Heat Balance

For Hot Fluid:

$Q_h = mC_p\Delta T$

$Q_h = \frac{5000kmol}{3600s} \times 128942$ J/kmol K × (472.28027 - 502) K

Q_h = - 5322390 W

For Cold Fluid:

$Q_c = mC_p\Delta T$

$Q_c = \frac{1556.2kmol}{3600s} \times 31190$ J/kmol K × (420 - 25.2443) K

Q_c = 5322390 W

LMTD:

"The log mean temperature difference is a logarithmic average of temperature difference between the hot and cold fluids at each end of a double pipe heat exchanger.**"**

The LMTD is used to determine the temperature driving force for heat transfer in flow system, mostly in heat exchangers.

Counter flow arrangement is selected as it is more efficient as compared to parallel flow because there is a uniform temperature difference between fluids over the whole length of the exchanger and fluid paths.

So, LMTD for counterflow arrangement is given as:

LMTD = $T_{lm} = \frac{\Delta Th - \Delta Tc}{\ln \Delta Th/\Delta Tc}$ = 215.24°C

For Heat Transfer Area:

$Q = U_D A \Delta T$

$A = Q/U_D \Delta T$

$U_D = 580\ W/m^2\ °C$ (Malaysia 2015)$A = 5322390\ W / (580\ W/m^2\ °C \times 215.24°C)$

$A = 42.63\ m^2 = 458.87\ ft^2 > (100 – 200ft^2)$
(KERN n.d.)

So, Heat Exchanger will be Shell & Tube type.

Also, Cold Fluid (air & n-butane) is in gaseous phase.

For Shell & Tube Heat Exchanger:

Heat Load = Q = 5322390 W

True Mean Temperature Difference (TMTD):

In most shell and tube heat exchangers, the flow will be a mixture of concurrent, countercurrent, and cross flow due to different shell-tube configurations. Therefore, the usual practice in the design of shell and tube heat exchanger is to estimate the "true temperature difference" from the logarithmic mean temperature by applying a correction factor "F_T" to allow for the departure from true countercurrent flow:

$TMTD = \Delta T_m = F_T \times \Delta T$ (here ΔT = LMTD)

F_T is a correction factor taken from the graph against the values of R & S.

Here, $R = (T_1 – T_2)/ (t_2 - t_1)$
(KERN n.d.)

R = 0.075

And, $S = (t_2 – t_1)/ (T_1 – t_1)$

S = 0.828

Therefore, $F_T = 0.93$
(KERN n.d.)

Now, $TMTD = F_T \times LMTD = 0.93 \times 215.24°C = 200.17°C$

Also, $F_T > 0.85$, so Take 1 Shell-Pass and 2-Tube Passes.

Therefore, A Simple 1-2 Heat Exchanger will be satisfactory.

Total Area (Required Heat Transfer Area):

$A = Q/U_D\Delta T$

Since $\Delta T = \Delta T_m$

$A= Q/U_D \Delta T_m$

$U_D = 580$ W/m^2 °C is taken as a trial value. Firstly, U_D was taken as 600W/m^2°C but the design was not satisfactory, therefore we now take $U_D = 580$ W/m^2 °C and proceed.

A= 5322390 W/ (580 W/m^2 °C × 200.17°C)

A = 45.84 m^2 = 493.42 ft^2

Fluid Allocation: (Shell & Tube side)

Corrosive/Toxic fluid is always kept on the tube side because it is cheaper to replace corroded tubes rather than the shell cover. Since none of the fluid in our case is corrosive/toxic so we allocate the molten salt to shell side because it is more viscous, has higher flow rate and has a large temperature change. Therefore, the other fluid (air-n-butane mixture) is allocated to the tube side.

Tube Parameters:

Tubes of Stainless Steel are selected, since they are stronger & highly corrosion resistive.

BWG = 16

O.D of tube = D_o = 1in.

I.D of tube = D_i = 0.870in.

Length = L = 16ft

Surface per lin. Ft = 0.2618 ft^2/ft.
(KERN n.d.)

Now, Number of Tubes are calculated as:

A = NaL

(KERN n.d.)

N= A/aL

N= 493.42ft^2/ (0.2618ft^2/ft. × 16ft)

N= 118 tubes.

Next Step is to assume pitch (it should always be greater than tube dia.).

We take square pitch as tubes are accessible for external cleaning and cause a lower pressure drop.

For O. D of tube = 1in, common pitch for square layout is 1 ¼ in. sq. pitch

Now, checking out for shell diameter with approximately equivalent no. of tubes as calculated.

The nearest count with 1shell & 2 tube passes with 1 ¼ sq. pitch:

N= 132 tubes

So, we proceed with N= 132 tubes the next nearest count with corresponding shell diameter D_s = 19.25in.

So, **Area w.r.t proposed geometry is:**

Corrected A_{real} = NaL

A_{real} = 132 × (0.2618 ft^2/ ft.) ×16ft

A_{real} = 553ft^2

Baffle Spacing:

Baffle Spacing can be taken 0.2 to 1 times of Ds (inside shell diameter), but the optimum spacing is in between 0.3 to 0.5 times Inside shell diameter.

(Gavin Towler n.d.)

On First Trial, Baffle Spacing was taken 0.5 times Ds, but the pressure drop on shell side was too high. (Not within allowable limits)

Therefore, Now taking baffle spacing = 0.6 × D_s

B= 11.55in.

4.1.3. Hot Fluid (Shell side) Calculations

- **Flow Area (a_s):**

Flow area is given as:

$as = (Ds \times C' \times B)/PT$

$a_s = (19.25in. \times 0.25in. \times 11.55in.)/(1.25in.)$

$a_s = 44.47in.^2$

$a_s = 0.31ft^2$

- **Mass Velocity (G_s):**

Mass Velocity is given as:

$Gs = m/as$

Mass flow rate = m= 391935 kg/hr. (Calculated from molar flow rate)

$Gs = (391935\ kg/hr)/0.31ft2$

$G_s = 1264306.4\ kg/hr.ft^2$

OR

$Gs = 2787795.6\ lb./hr.ft^2$

Shell-side Equivalent Diameter (D_e):

$D_e = 4\ [PT\ 2 - (\pi do\ 2)/4]/\ (\pi do)$

$D_e = \frac{4[\left(1\frac{1}{4}in\right)^2 - \frac{\pi(1in)^2}{4}]}{\pi(1in)}$

$D_e = 0.989$ in.

$D_e = 0.0824ft$

Since Molten Salt ($NaNO_2$ & $NaNO_3$ mixture) is a viscous fluid but on the tube side there is a gaseous mixture. Therefore, the properties of both fluids are evaluated at their mean temperatures.

$T_{AVG.}$ For Hot Fluid = 487.1°C

At 487.1°C,

Viscosity,

μ = 1.3104 cP
(DON W. GREEN n.d.)

μ = 3.16997 lb./ft.h.

Thermal Conductivity, k:

k = 115.741 J/ h.ft.K.
(DON W. GREEN n.d.)

Heat Capacity,

C_P = 128134J/ kmol. K.
(DON W. GREEN n.d.)

- **NR_e** = $D_e \times G_s / \mu$

NR_e = 0.0824ft × (2787795.6 lb./h.ft^2)/ (3.16997 lb./ft.h.)

NR_e = 72466

Heat Transfer Coefficient (h_o):

For values of R_e from 2000 to 1,000,000, the data is closely represented from the equation: (KERN n.d.)

$$ho\ De\ /k\ =\ 0.36\ (De\ Gs\ /\mu)0.55\ \times (c\mu/k)1/3\ \times\ (\mu/\mu w\ )0.14$$

$h_o\ D_e/k = 0.36 \times (72466)^{0.55} \times (20.31)^{1/3} \times (1)^{0.14}$

Since viscosity factor (μ/μ_w) is considered equal to 1.

$h_o\ D_e/k$ = 462.67

Substituting D_e and k, gives h_o:

h_o = 649877.3 J/h ft^2 K

4.1.4. Cold Fluid (Tube side) Calculations

Flow area (a_t):

Flow area per tube = a_t'= 0.594in^2
(KERN n.d.)

Total Flow area of tubes = a_t = N (a_t')/n

a_t = [132×0.594in.^{2}]/2

a_t = 39.204in.^{2} = 0.272ft^2

Mass Velocity (G_t):

G_t = m/a_t

G_t = (45870.8kg/hr.)/0.272ft^2

G_t = 168643 kg/hr.ft^2

OR

G_t = 371858 lb./hr.ft^2

Since Molten Salt ($NaNO_2$ & $NaNO_3$ mixture) is a viscous fluid but on the tube side there is a gaseous mixture. Therefore, the properties of both fluids are evaluated at their mean temperatures.

$T_{AVG.}$ For Cold Fluid = 222.62°C

At 222.62°C,

Viscosity,

μ = 0.064296 lb/ft.h (the NIST WebBook n.d.)

Thermal Conductivity, k:

k = 43.01 J/h. ft. K (the NIST WebBook n.d.)

Heat Capacity,

C_P = 31.96 J/mol. K (the NIST WebBook n.d.)

NR_e = $D_i \times G_t / \mu$

D_i = 0.0725ft.

NR_e = 0.0725ft × (371858 lb./h.ft^2)/ (0.064296 lb./ft.h)

NR_e = 419306

Heat Transfer Coefficients (h_i & h_{io}):

The general equation that can be used for exchanger design is:

$Nu = C \times R_e^{0.8} \times Pr^{0.33} \times (\mu/\mu_w)^{0.14}$
(Gavin Towler n.d.)

Since, our cold fluid are gases, so for gases C is taken as 0.021.
(Gavin Towler n.d.)

$(\mu/\mu_w)^{0.14}$ is viscosity factor which is considered equal to 1.

R_e = 419306

$Pr = (C\mu/k) = 0.735$

Nu = 597.2

$Nu = (h_i\, d_i)/ k$

So,

$h_i = 597.2 \times k/ d_i$

h_i = 354284 J/h.ft^2. K

Also,

$h_{io} = h_i \times (I.D)/ (O.D)$

h_{io} = 354284 J/h.ft^2. K × (0.0725ft)/ (0.0833ft)

h_{io} = 308350.42 J/h.ft^2. K

Overall Coefficient:

Firstly, calculate Clean Overall Coefficient (U_c):

$U_c = \frac{hio \times ho}{hio + ho}$

U_c = 209125.6 J/hr.ft^2. K

4.1.5. Calculations for U_D

$\frac{1}{Ud} = \frac{1}{Uc} + R_d$

Using Fouling factor, 0.0002 m^2 °C/W for both fluids.

(Malaysia 2015)

$\frac{1}{Ud} = \frac{1}{209125.6J/(3600s)(0.0929m2)°C}$ + (0.0002 +0.0002) (m^2 °C/ W)

U_D = 507.6 W/m^2 °C

Now, Calculated Area = Q/U_D ΔT

$A_{CALCULATED}$ = 52.4 m^2

Also, A_{REAL}= NaL= 51.4 m^2

Before, proceeding to Pressure Drop Calculations, A_{REAL} = $A_{CALCULATED}$.

Here, A_{REAL} ≈ $A_{CALCULATED}$.

So, we go for pressure drop calculations.

4.1.6. Pressure Drop Calculations

Shell side (Hot Fluid) Pressure drop:

It is given by the formula:

$\Delta Ps = \frac{f\ Gs^2\ (Nb+1)Ds}{2g\rho De\Phi s}$

For f:

NRe = 72466 > 20000 so,

f = 0.184 (NRe) ^ (-1/5)

f= 0.0196

Mass Velocity = G_s = 2787795.6 lb./hr.ft^2

No. of Crosses = 12(L/ B) = 17

Ds = 19.25in. = 1.604ft

g = 32.17 ft^2 /s

ρ at Tavg. = 120.3 lb. /ft^3

So,

ΔPs = 3.48 psi < 10psi (Allowable)

Tube Side (Cold Fluid) Pressure drop:

For f:

NRe = 419306 > 20000

So, f = 0.184 $(NRe)^{-1/5}$

f = 0.014

I.D = Di = 0.0725ft

g = 32.17 ft. / s^2

Density at $T_{avg.}$ = $\rho_{avg.}$ = 0.1249 lb. / ft^3
(the NIST WebBook n.d.)

Pressure drop is given by:

$$\Delta Pt = \frac{f\ Gt^2 Ln}{2\rho g Di \Phi}$$

So, by putting values we get:

$\Delta P_t = (1.063 \times 10^9) / (3600)^2 \times 144$

ΔP_t = 0.569 psi

Also,

$\Delta Pr = 4n\ Gt^2 / 2g\rho\Phi$

By, putting values we get:

$\Delta P_r = (1.37 \times 10^8) / (3600)^2 \times 144$

$\Delta P_r = 0.73$ psi

Total Pressure Drop on tube side = $\Delta P_T = \Delta Pt + \Delta Pr = 1.29$ psi < 10 psi, so within allowable limits.

4.2. Reactor Design

4.2.1. Reactor Selection

Catalytic Fixed bed reactors are one of the most important reactors for producing abundant amount of chemical and intermediates to produce maleic anhydride. Let's make it easy by eliminating the reactors. Batch and CSTR is automatically removed due to the gas phase reactions next PFR, but it is also eliminated because we have catalyst involved so, the last option is PBR.

In PBR we have two options Fluidized Bed and Multi tubular Fixed Bed reactor.

There is one main concern the dissipation of heat since we have an exothermic reaction and to maintain the temperature. The dissipation of heat is best in fluidized bed, but the biggest problem is we cannot control the temperature within the reactor but in multi tubular fixed bed reactor we can control and dissipate the heat using interchanging heat exchangers.

4.2.2. Chemical Reaction

$$C_4H_{10} + 3.5O_2 \rightarrow C_4H_2O_3 + 4H_2O$$

Table 4.1: Mass Balance around Reactor

Components	**Molar flow Inlet (F_{IO}) kmol/h.**	**Molar fraction Inlet kmol/h.**	**Molar flow Outlet kmol/h.**	**Molar fraction Outlet` kmol/h.**
CO_2 =	19.0708233	0.00645543	43.06573637	0.014450098
O_2 =	550.6700227	0.18640049	392.7035116	0.131766098
N_2 =	2337.538055	0.79125104	2337.538055	0.78432777
H_2O =	0	0	165.9648155	0.055687142
C_4H_{10} =	46.95170771	0.01589304	7.04273916	0.002363091
MAN =	0	0	33.99279353	0.0114058
Total	2954.230609	1	2980.307651	1

4.2.3. Composition

$$C_{TO} = \frac{Po}{R*To}$$ (H. Scott Fogler 1999)

$$= \frac{270\ \text{Kpa}}{8.314\ \frac{\text{KJ}}{\text{Kmol} * \text{K}} * \frac{693\ \text{K}}{1}}$$

$= 0.046861967\ \frac{\text{Kmol}}{\text{m}^3}$

$C_{AO} = Y_{ao} * C_{TO}$

$= 0.01589304 * 0.04686196$

$= 0.000744779\ \frac{\text{Kmol}}{\text{m}^3}$

$\varepsilon = (4+1-3.5-1) * Y_{ao}$

$\varepsilon = (4+1-3.5-1) * 0.01589304$

$\varepsilon = 0.0079$

$k_1 = (0.96*10^{\wedge}(-6))$ kmol (kg^{-1}) (sec^{-1}) $(atm^{0.54})$

(Patrick V. Mangili, Diego M. Prata 2020)

$k_1 = 0.041846651$ kmol.kg^{-1}.h^{-1}. $kpa^{0.54}$

$K_2 = 3.059462127$ kpa-1

Table 4.2: Concentration of the reactant and products

1+εX	1	1.0019866	1.003973	1.00596	1.007152
X	0	0.25	0.5	0.75	0.9
C_{AO} (kmol/m3)	0.000744779				
C_A (kmol/m3)	0.000677749	0.0005073	0.000338	0.000168	6.73E-05
C_B (kmol/m3)	0.007948935	0.0073413	0.006736	0.006133	0.005773
C_C (kmol/m3)	0.000490687	0.0006588	0.000826	0.000993	0.001093
C_D (kmol/m3)	0.002395706	0.0030674	0.003736	0.004403	0.004801

Now,

$C_A = CAO * (1 - X) * \frac{1}{(1+\varepsilon)} * Y$ [Y = P/Po]

$C_B = \left(\frac{CAO*Y*\left(\theta B - \left(\frac{b}{a}\right)*X\right)}{(1+\varepsilon X)} \right)$ [Where $\Theta_B$is F_{Bo}/F_{AO}, b is the coefficient of O_2 which is 3.5]

$C_C = \left(\frac{CAO*Y*\left(\theta C - \left(\frac{c}{a}\right)*X\right)}{(1+\varepsilon X)} \right)$

$C_D = \left(\frac{CAO*Y*\left(\theta D - \left(\frac{d}{a}\right)*X\right)}{(1+\varepsilon X)} \right)$ [Where d is 4]

Now,

P_O = 270 KPa

L = 4.2 (Patrick V. Mangili, Diego M. Prata 2020)

Now we will calculate the volume of the reactor.

$R = \frac{k1*Pa^{\alpha}}{(1+K2*Pc)}$ (Ramesh K. Sharma and David L.Cresswell 1991)

$P_a = (C_A*R*T)$

$C_A = \frac{CAo*(1-X)*Y}{(1+\varepsilon)}$

$C_C = \left(\frac{CAO*Y*\left(\theta C - \left(\frac{c}{a}\right)*X\right)}{(1+\varepsilon X)} \right)$

$$R = \frac{0.041846651*\left(\frac{CAo*(1-X)*Y*R*T}{(1+\varepsilon)}\right)^{\alpha}}{\left(1+K2*\left(\frac{CAO*Y*\left(\theta C - \left(\frac{c}{a}\right)*X\right)*R*T}{(1+\varepsilon X)} \right)\right)}$$

Now for X= 0

$$R = \frac{0.041846651*\left(\frac{0.000744779*(1-0)*.91*8.314*693}{(1)}\right)^{.54}}{\left(1+3.059462127*\left(\frac{0.000744779*.91*\left(0.72 - \left(\frac{1}{1}\right)*(0)\right)*8.314*693}{1} \right)\right)}$$

R = 0.009049507

Now,

Table 4.3: Finding rate (R) from Conversion (X)

Conversion X	Rate (R)	(1/R)
0	0.009049507	110.5032531
0.25	0.005920707	168.8987451
0.5	0.003850278	259.7215158
0.75	0.002224981	449.4419737
0.9	0.001238039	807.7288826

4.2.4. Graph for 90% Conversion

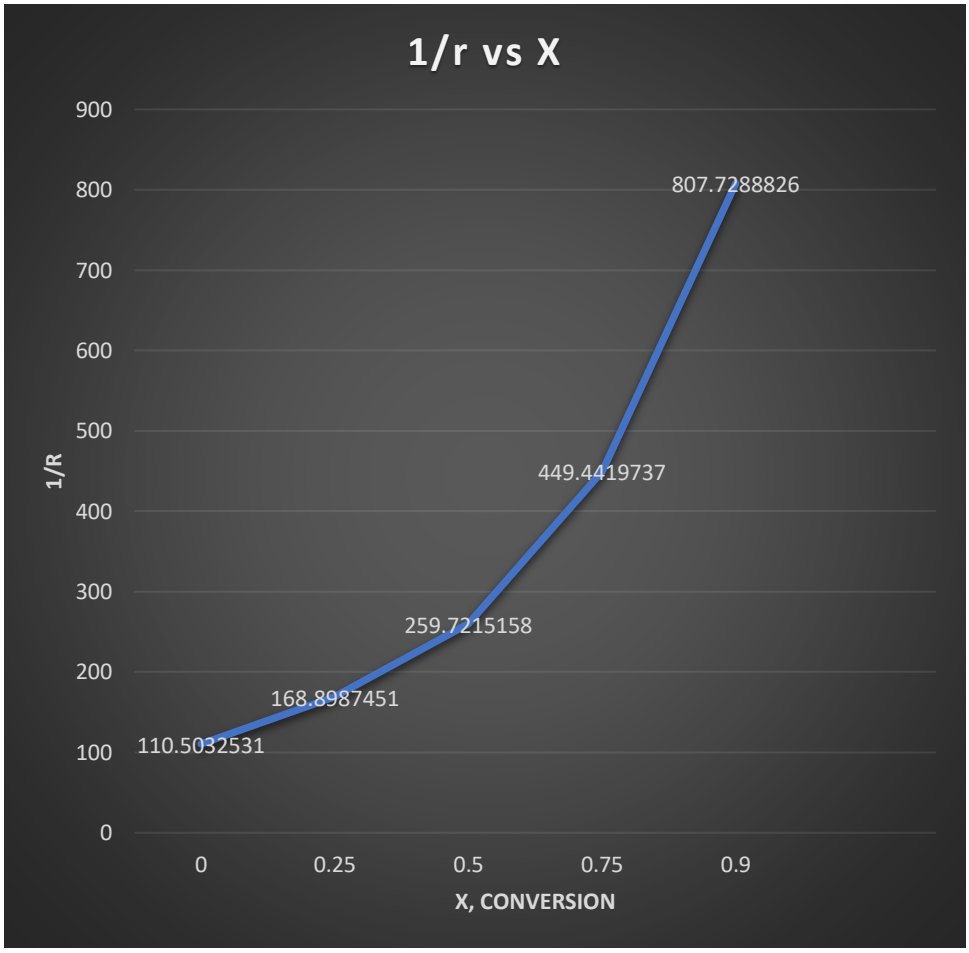


Figure 4.1: Graph of 1/Rate (1/R) vs. Conversion (X)

4.2.5. Calculations for volume

$$V = \left(\frac{0.25*46.95}{3} * [110.5032531 + 168.8987451 + 259.721515 + 449.4419737 + 807.7288826]\right)$$

$$V = 7028.257353 \frac{Kg}{1}$$

$$V = \frac{7028.257353}{2000} * \frac{\cancel{Kg}*m^3}{\cancel{kg}}$$

$$V = 3.514128676\ m^3$$

This volume is calculated for 90 percent conversion

Now calculating for 85 percent conversion.

4.2.5. Graph for 85% conversion

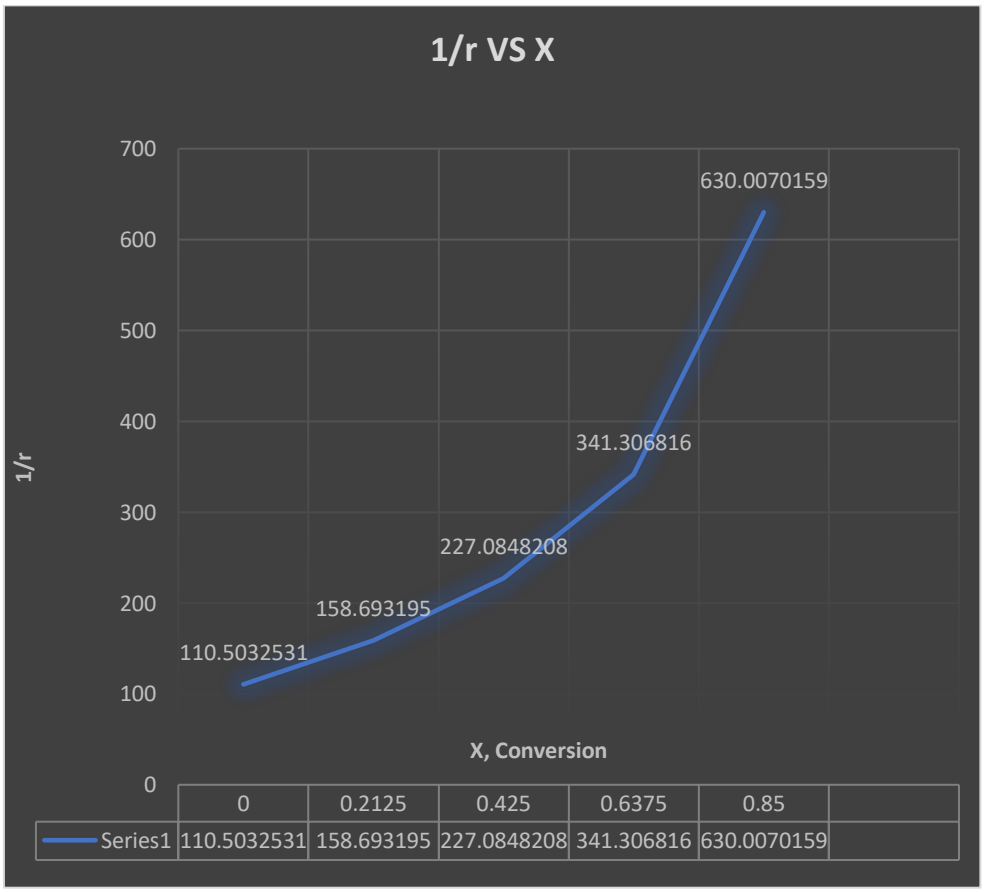


Figure 4.2: Graph of 1/Rate (1/R) Vs Conversion (X)

4.2.6. Calculations of volume

$$V = \left(\frac{0.2125*46.95}{3} * [110.5032531 + 158.693 + 227.085 + 341.307 + 630.007]\right)$$

$$V = 5742.17 \ \frac{Kg}{1}$$

$$V = \frac{5742.17}{2000} * \frac{\cancel{Kg}*m^3}{\cancel{kg}}$$

$$V = 2.87 \ m^3$$

$$V = 2.9 \ m^3$$

4.2.7. Calculation of Weight of Catalyst

The weight of the catalyst is an important feature of the packed bed or fixed bed reactor. The catalyst used in this reactor is VPO also known as “Vanadium Phosphorus Oxide “There were other options as well i.e. “Vanadyl pyrophosphate (VO) 2P2O7” there is huge debate in which catalyst to choose for but this debated was ended due to help of an article which concluded that VPO is more selective maleic anhydride than any catalyst. (Y. Zhanglin, M. Forissier, R.P. Sneeden, J.C. Vedrine, J.C. Volta 1994)

```
b = ((Cao*(0.723994828+(X))*RTY)/(c))
c = (1+(epi*X))
d(y) / d(W) = (-(alpha*(c)))/(2*y)
y(0) = 1
d(X) / d(W) = (k1*a)/(1+(k2*b))
X(0) =0
W(0) = 0
W(f) = 450
k2 = 3.059462127
k1 = 0.041846651
alpha = 1.94*10^(-4)
epi = 0.0079
T = 693
Cao = 0.000744745
R = 8.314
a = ((RTY*(Ca))/(c))^(0.54)
Ca = Cao*(1-X)
RTY = R*T*y
```

Figure 4.3: Screenshot of Inserting Data in Polymath workspace

W	y	X
181.9366	0.9821276	0.758798
185.5366	0.9817699	0.7662898
189.1366	0.981412	0.7736209
196.3366	0.9806957	0.7878133
199.9366	0.9803373	0.7946804
203.5366	0.9799788	0.8013985
207.1366	0.9796201	0.8079701
214.3366	0.9789024	0.8206841
217.9366	0.9785432	0.8268312
221.5366	0.978184	0.8328415
225.1366	0.9778246	0.838717
232.3366	0.9771053	0.8500726
235.9366	0.9767454	0.8555566
239.5366	0.9763854	0.8609139

Figure 4.4: Screenshot from Polymath representing the weight of catalyst at 85% conversion

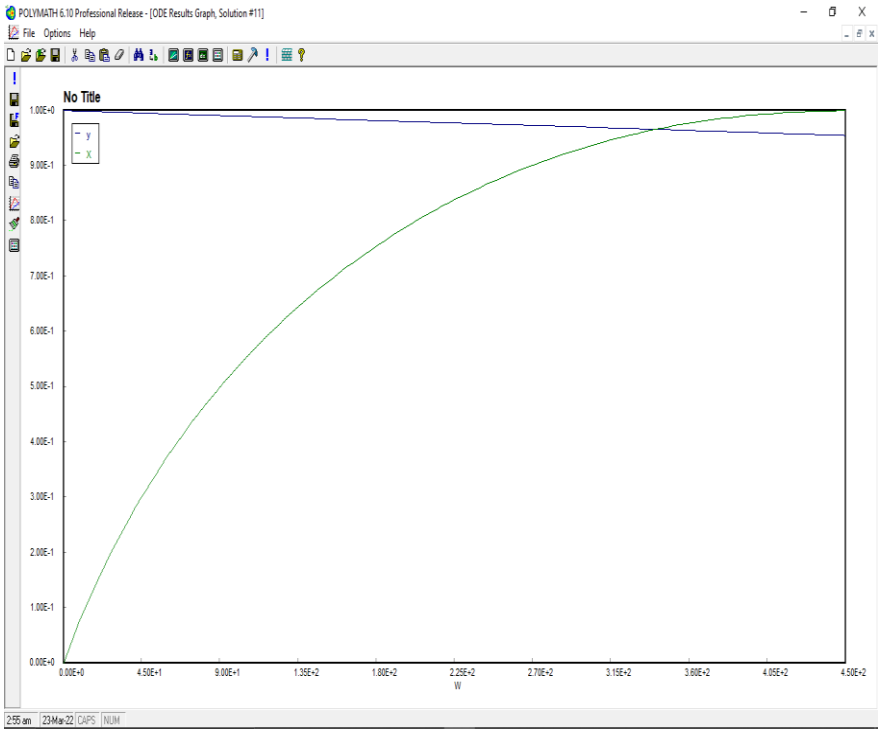


Figure 4.5: Screenshot from Polymath representing the trend of Pressure ratio (Y) with Weight of Catalyst (W)

Calculated values of DEQ variables

	Variable	Initial value	Minimal value	Maximal value	Final value
1	a	2.195723	0.0102898	2.195723	0.0102898
2	alpha	0.000194	0.000194	0.000194	0.000194
3	b	3.106607	3.106607	7.021868	7.009699
4	c	1.	1.	1.0079	1.0079
5	Ca	0.0007447	3.82E-08	0.0007447	3.82E-08
6	Cao	0.0007447	0.0007447	0.0007447	0.0007447
7	epi	0.0079	0.0079	0.0079	0.0079
8	k1	0.0418467	0.0418467	0.0418467	0.0418467
9	k2	3.059462	3.059462	3.059462	3.059462
10	R	8.314	8.314	8.314	8.314
11	RTY	5761.602	5502.829	5761.602	5502.829
12	T	693.	693.	693.	693.
13	W	0	0	450.	450.
14	X	0	0	0.9999487	0.9999487
15	y	1.	0.9550865	1.	0.9550865

Differential equations

1 d(y)/d(W) = (-(alpha*(c)))/(2*y)
2 d(X)/d(W) = (k1*a)/(1+(k2*b))

Explicit equations

1 R = 8.314
2 epi = 0.0079
3 k2 = 3.059462127
4 k1 = 0.041846651

Figure 4.6: Screenshot of summary from Polymath

From the above we came to an understating that for maximum of 0.85 conversion the amount of catalyst required in terms of Kg is 232.3366 Kg and other than that the total pressure drop other than eurgon equation is 0.97 which is less than that calculated in ergun equation.

4.3. Design of Distillation Column

4.3.1. Distillation

The separation of the components of a mixture, carried out based on difference in their boiling points is known as Distillation. It is Physical separation process, as no Chemical Reaction is involved in the separation. Different types of distillation processes are carried out on industrial scale, some of them are discussed as:

- Simple Distillation (Suitable when minimum ΔT_b of the components is 25^0C).
- Vacuum Distillation (Suitable when the Boiling point of the Solvent is high i.e., higher than 150^0C).
- Fractional Distillation (Suitable when the boiling point difference between the components is smaller i.e., less than 25^0C)
- Steam Distillation (Suitable for heat sensitive compounds i.e., heating and separation of the compounds at temperature below their decomposition point)
- Azeotropic Distillation (Suitable for Azeotropic Mixtures i.e., having components with almost same boiling points, on boiling the vapors have same composition as in the Liquid)

Since the Mixture in this case, contains the components:

Water (solvent) with a $T_b = 100^0$C, & Maleic Anhydride (Solute or Product) with $T_b = 202^0$C. As the boiling point of solute is less than 150^0C, the temperature difference between the components is higher 25^0C and the mixture is not an azeotropic mixture. So, the distillation process selected in this case is Simple Distillation.

4.3.2. Types of Distillation Column

The distillation process is carried out in a Distillation Column. Mainly, there are two types of Distillation Columns:

4.3.2.1. Packed Columns

The working of packed column is based on the continuous Liquid-Vapor contact throughout the packed bed. The Liquid flows from the to the bottom of the column and gas flows counter-currently in the direction of the liquid flow. The performance of a packed column mainly depends upon the good Liquid-Vapor distribution and contact in throughout the Packed bed. The Packed Columns mainly contains a liquid distributor,

a hold down plate and the Packing support. The packing's used in the Packed column are of two different types i.e.,

1. Packings with regular geometry.
2. Packings with the random geometry.

4.3.2.3. Tray Columns

These columns consist of trays with suitable tray spacing (range between 0.15m -1m). The height of the Distillation column depends upon the plate spacing. The plate spacing depends upon the diameter of the column and the operating conditions. A tray column mainly uses three types of trays:

1. **Sieve-plates** (the vapor passes through the holes and the liquid is retained on the plate, the liquid starts weeping through the holes at low flowrates, hence reducing the plate efficiency).
2. **Bubble-cap Plates** (the vapor passes through small pipes, which are called risers, covered with caps such that there is a slot between the riser wall and the edge of the cap which allow the vapors to flow. This design has a good turn down ratio even at low flowrates, high liquid-vapor Contact time).
3. **Valve-Plates** (Sieve Plates with movable flaps which lift according to flow of the vapors through the holes. They offer less contact time than the bubble-cap plates, pressure drop is high).

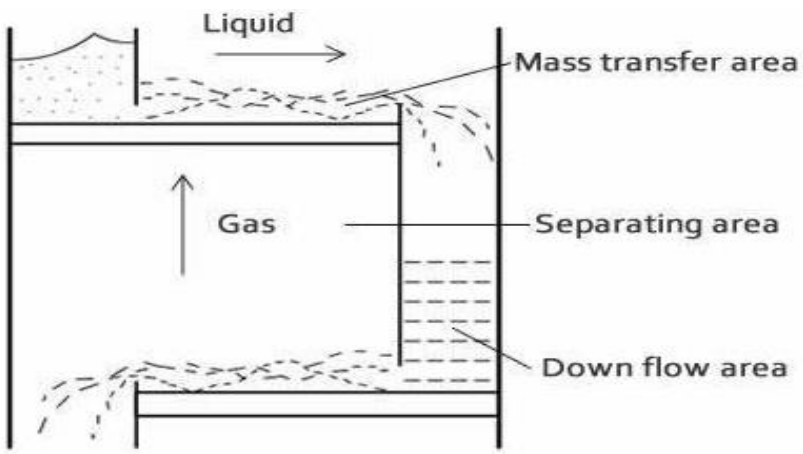


Figure 4.7: Working Principle of a Tray Column

In this case, a tray column is selected as the diameter of the column is higher than 0.6m and the distillation column is designed for a toxic and corrosive application. Moreover, the bubble-cap trays are selected as they offer high liquid-vapor contact time hence maximum mass transfer will be achieved with least weeping.

4.3.3. Design of the Equipment

Table 4.4: Feed rate & composition of the Distillation Column

Components	**In Feed**		**In Top**		**In Bottom**	
	Flow rate (Kg/h)	**Mass Fraction**	**Flow rate (Kg/h)**	**Mass Fraction**	**Flow rate (Kg/h)**	**Mass Fraction**
Water	5430.876	0.62	5426.22	0.998	6.648	0.002
MAN	3328.602	0.38	10.87	0.002	3317.73	0.998
Total	8759.479	1	5435.099	1	3324.38	1

So, there must be 99.8% water in Top Product and not more than 0.2% water in Bottom Product, which is Maleic Anhydride.

Hence,

X_D =0.998

X_w = 0.002

4.3.4. Temperature of the Feed

The Feed of distillation column is at its boiling point which is calculated as:

Weight Average Boiling Point = $\sum xi\ Tbi$

Where,

x_i = Mass fraction of the components in Feed.

Tb_i = Boiling Point Temperature of the components in the Feed.

Boiling Point of Maleic Anhydride @ 1 bar = 202 ^{0}C

Boiling Point of Water @ 1 bar = 100 ^{0}C

As,

$W_{ABP} = x_{H2O}\ Tb_{H2O} + x_{MAN}\ Tb_{MAN}$

So,

W_{ABP} = (0.62*100) + (0.38*202) = 138.8 ^{0}C @ 1 Bar.

4.3.5. Calculation of Relative Volatility

Relative Volatility can be calculated by the following formula:

Relative Volatility (α) = P_A^o/P_B^o ____ (A) (richardson and coulson 2002)

Where,

P_A^o is Vapor Pressure of Water at Feed Temperature i.e., 138.8 $^\circ$C,

P_B^o is Vapor Pressure of Maleic Anhydride at Feed Temperature i.e., 138.8 $^\circ$C.

Calculating the vapor Pressure by using Antoine's Equation:

Log_{10} (P) = A - (B / (T + C))

Where,

P is the vapor pressure (in bars) of the components at the Temperature (T in K)),

A, B & C are constants taken from the NIST Web Book.

For component A @ 411.8K:

For Temperature range: (379K – 573K)

A = 3.55959,

B = 643.748,

C = -198.043.

By putting values in the Antoine's Equation, we get:

P_A^o = 3.5318.

For component B @ 411.8K:

For Temperature range: (317K – 475K)

A = 3.79916,

B = 1431.009,

C = -101.093.

By putting values in the Antoine's Equation, we get:

$P_B^o = 0.1561$.

By equation (A):

Relative Volatility (α) = 22.62

Relative Volatility is also calculated by the following relation:

$\alpha = (y_A\ x_B) / (y_B\ x_A)$ ____ (B)

Where,

y_A is the mass fraction of water in vapor Phase,

x_A is the mass fraction of water in liquid Phase,

y_B is the mass fraction of Maleic Anhydride in vapor Phase,

x_B is the mass fraction of Maleic Anhydride in liquid Phase.

For liquid composition:

As,

$x_A = (P - P_B^o) / (P_A^o - P_B^o)$

$x_A = 0.2499$

Also,

$x_A + x_B = 1$

So,

$x_B = 1 - x_A$

$x_B = 0.7501$.

For vapor composition:

As,

$y_A = (P_A^o\ x_A) / P$

$y_A = 0.883$

Also,

$Y_B = (P_B^o\ x_B) / P$

$Y_B = 0.1170$

Hence, from equation (B):

$\alpha = 22.65$

4.3.6. Calculation of Minimum Reflux Ratio

$R_{min} = \frac{1}{(\alpha - 1)} * (\frac{xDA}{xfA} - \frac{\alpha(1-xDA)}{(1-xfA)})$ (richardson and coulson 2002)

Putting the values in the equation:

$R_{min} = \frac{1}{(22.65 - 1)} * (\frac{0.998}{0.62} - \frac{22.65(1-0.998)}{(1-0.62)})$

$R_{min} = 0.0688$

4.3.7. Calculation of Actual Reflux Ratio

$R_{Actual} = 1.3 * R_{min}$

$R_{Actual} = 0.0894$

4.3.8. Calculation of Minimum Number of stages

By Fenske's Equation:

$N_{min} = \left(\frac{\ln\left(\frac{xd(1-xw)}{xw(1-xd)}\right)}{\ln(\alpha)}\right) - 1$ (Dutta 2009) (richardson and coulson 2002)

$N_{min} = \left(\frac{\ln\left(\frac{0.998(1-0.002)}{0.002(1-0.998)}\right)}{\ln(22.65)}\right) - 1$

$N_{min} = 2.981$ equivalent to 3 plates.

Now, applying the total balance on Distillation Column:

F = D +W

Feed flowrate = 8759.479 kg/h

Required Flow rate of the Bottom product W = 3333.33 kg/h___ (1)

Putting value of eq (1) in total balance equation:

F – W = D (richardson and coulson 2002)

8759.479 – 3333.33 = D

D = 5426.149 kg/h.

By using the following relation:

$R_{Actual} = L_n / D$

L_n = 485.09 kg/h.

By using the following relation:

$V_n = L_n + D$

V_n = 5911.239 kg/h.

4.3.9. Top Operating Equation

$y_n = \frac{Ln}{Vn}(xn + 1) + \frac{D}{Vn}xd$

$y_n = \frac{485.09}{5911.239}(xn + 1) + \frac{5426.149}{5911.239}(0.998)$

y_n = 0.0820 (xn+1) + 0.9161 _____ (2)

Using the following relation:

$L_m = L_n + F$

L_m = 9244.569 kg/h.

Using the following relation:

$V_m = L_m – W$

V_m = 5911.239 kg/h.

4.3.10. Bottom Operating Equation

$y_m = \frac{Lm}{Vm}(xn + 1) - \frac{W}{Vm}xw$

$y_m = \frac{9244.569}{5911.239}(xn + 1) - \frac{3333.33}{5911.239}(0.002)$

y_m= 1.563 (xn+1) - 0.00112 ____ (3)

$x = \frac{y}{\alpha-(\alpha-1)y}$ _____ (4)

4.3.11. Calculation of Number of ideal stages using Lewis-Sorel method

Using the equations (2), (3) and (4) to find the liquid and vapor composition on each stage:

At y_t = 0.998

Using the equation (4): x_t = 0.9565.

First, using top operating equation for stages above the feed point:

Using the equation (2): y_{t-1} = 0.9945

Using the equation (4): x_{t-1} = 0.8893

Using the equation (2): y_{t-2} = 0.9890

Using the equation (4): x_{t-2} = 0.7991

Using the equation (2): y_{t-3} = 0.9816

Using the equation (4): x_{t-3} = 0.7022

Using the equation (2): y_{t-4} = 0.9736

Using the equation (4): x_{t-4} = 0.6203

Using the equation (2): y_{t-5} = 0.9669

Using the equation (4): x_{t-5} = 0.5637

This value is close to the fraction in feed i.e., x_{fH20} = 0.62. So, the feed is introduced on the plate “t-5”.

Now, using the Bottom Operating equation (3) to find the fractions on the stages below the Feed point.

Using the equation (3): y_{t-6} = 0.8799

Using the equation (4): x_{t-6} = 0.2444

Using the equation (3): $y_{t-7} = 0.3808$

Using the equation (4): $x_{t-7} = 0.0264$

Using the equation (3): $y_{t-8} = 0.0402$

Using the equation (4): $x_{t-8} = 0.0018$

Xt-10 is slightly weaker than the minimum required fraction of Water in Bottom Product i.e., $x_{wH2O} = 0.002$. So, the Bottom product is withdrawn. Hence, t-8 is the reboiler and the total number of ideal stages is 8 with feed plate "t-6".

4.3.12. Calculation of Column Efficiency & Actual Trays

Finding the value of Column Efficiency from Figure 11.13 in Coulson & Richardson's Chemical Engineering, Volume 6, (R K Sinnott 2005)

by using the value of $\mu_a\alpha_a$:

Where,

μ_a is the Molar Average Viscosity,

α_a is the Average Relative Volatility of lighter component.

As,

$\alpha_a = y_a / x_a$

$y_a = 0.883$, $x_a = 0.2499$

So,

$\alpha_a = 3.5334$

Column Temperature = (99.6 + 201.5) / 2 = 150.65 ^{0}C.

$\mu_{H2O} = 0.1826$ @ 150.65 ^{0}C.

$\mu_{MAN} = 0.61$ @ 150.65 ^{0}C.

In Feed molar Fractions are:

$x_{H2O} = 0.899$,

$x_{MAN} = 0.101$

Hence,

$\mu_a = (x_{H2O}\ \mu_{H2O}) + (x_{MAN}\ \mu_{MAN})$

$\mu_a = 0.2257$

And

$\mu_a\ \alpha_a = (0.2257*3.5334)$

$\mu_a\ \alpha_a = 0.7977$

Now, finding the value of Column Efficiency from the graph.

$E_o = 54\ \%$

Number of Actual Plates = N_{ideal} / E_o

Hence,

Number of Actual Plates = 14.81 equivalent to 15 Actual Trays

4.3.13. Feed Point Calculation

The Feed-point can be determined by using the Kirkbride Equation:

$$\text{Log}\left(\frac{Nr}{Ns}\right) = 0.206\text{Log}\left(\frac{B}{D} * \frac{xf,MAN}{xf,H2O} * \left(\frac{xw,H2O}{xD,MAN}\right)^2\right)$$ (R K Sinnott 2005)

$$\text{Log}\left(\frac{Nr}{Ns}\right) = 0.206\text{Log}\left(\frac{3333.33}{5426.149} * \frac{0.38}{0.62} * \left(\frac{0.002}{0.002}\right)^2\right)$$

Hence,

$\frac{Nr}{Ns} = 0.817$

$N_r = 0.817 * N_s$

Also,

$N_r + N_s = N_{Actual}$

$1.817\ N_s = 15$

So,

$N_s = 8.25$ equivalent to 8 Trays,

$N_r = N_{Actual} - N_s = 7$ Trays.

Hence, there are 8 Trays below the Feed point and 7 trays above the feed Point, whereas the feed point is the 8th tray from the Bottom.

4.3.14. Mechanical Design

Data for Design:

Theoretical Plates (ideal stages) = 8 plates

Column Efficiency = 54%

Slope of Top Operating Line = $\frac{Ln}{Vn} = \frac{485.09}{5911.239} = 0.0820$

Slope of Bottom Operating Line = $\frac{Lm}{Vm} = \frac{9244.569}{5911.239} = 1.563$

4.3.15. Liquid and Vapor Densities of Top and Bottom of the column

Average Molecular Weight: $\sum xi * (M.wt.i)$

Where,

xi is the molar fraction of the components in the mixture,

M.wt.i is the molecular weight of the components.

Mol. $Wt._{Avg}$ for Top: (0.000367*98.06) + (0.9996*18) = 18.028 g/mol.

Mol. $Wt._{Avg}$ for Bottom: (0.0107*18) * (0.9891*98.06) = 97.183 g/mol.

Vapor Density at the Top of the column:

$\frac{PM}{RT} = \frac{(101325\ Pa)(18.028\frac{g}{mol})}{(8.314\frac{m3*Pa}{K*Mol})(372.6K)}$ = 589.673 g/m^3 or 0.5896 kg/m^3.

Vapor Density at the Bottom of the column:

$\frac{PM}{RT} = \frac{(101325\ Pa)(97.183\ \frac{g}{mol})}{(8.314\frac{m3*Pa}{K*Mol})(474.5\ K)}$ = 2496.092 g/m^3 or 2.4961 kg/m^3.

Liquid Density:

Density of liquid = $\sum xi\ \rho i$

Where,

xi is the mass fraction of the components in the mixture,

ρi is the density of the components w.r.t the temperature of the mixture.

Liquid Density at top of the column:

Density of liquid: (0.998*958.676) + (0.002*1121) = 956.00 kg/m^3.

Liquid Density at Bottom of the column:

Density of liquid: (0.002*861.47) + (0.998*663.2) = 665.59 kg/m^3.

Liquid surface Tension at the Top of the column (σ) = 58.3054 mJ/m^2.

Liquid surface Tension at the Bottom of the column (σ) = 23.3851 mJ/m^2.

Range of plate spacing = 0.15m to 1m

4.3.16. Calculation of the Diameter of the column

F_{LV} for Top:

$$F_{LV} = \frac{Ln}{Vn}\sqrt{\frac{\rho v}{\rho L}}$$

$$=\frac{485.09}{5911.239}\sqrt{\frac{0.5896}{956.00}}$$

= 0.002037

F_{LV} for Bottom:

$$F_{LV} = \frac{Lm}{Vm}\sqrt{\frac{\rho v}{\rho L}}$$

$$=\frac{9244.569}{5911.239}\sqrt{\frac{2.4961}{665.59}}$$

=0.09577

Taking Tray spacing of 0.5m and by using the Figure 17.34 (Book: Chemical Engineering design by Galvin Towler & Ray Sinott)
(Sinnott, Galvin Towler and Ray 2002)

finding the value of K_1:

K_1 for Top = 0.05

K_1 for Bottom = 0.081

Corrected values of K_1:

$K_{1C} = K_1 * (\frac{\sigma}{0.02})$^0.2

For Top:

$K_{1C} = 0.077 * (\frac{0.583054}{0.02})$^0.2

$K_{1C} = 0.098$

For Bottom:

$K_{1C} = 0.083 * (\frac{0.233851}{0.02})$^0.2

$K_{1C} = 0.1357$

4.3.16. Calculation of Flooding Velocity

$\mu f = K_{1C}\sqrt{\frac{\rho L - \rho V}{\rho V}}$

For Top:

$\mu f = 0.098 * \sqrt{\frac{959.0 - 0.58967}{0.58967}}$

$\mu f = 3.95$ m/s

For Bottom:

$\mu f = 0.1357\sqrt{\frac{663.59 - 2.496}{2.496}}$

$\mu f = 2.208$ m/s

4.3.17. Design Velocity

For Design, the range of velocity is 80% to 85% of the flooding velocity is used (Chapter 17, Book: Chemical Engineering design by Galvin Towler & Ray Sinnott)

So, the Design velocity is 85% of the flooding velocity:

Design Velocity = 0.85 * Flooding Velocity

For Top:

Design Velocity = 0.85 * 3.95

= 3.3575 m/s

For Bottom:

Design Velocity = 0.85 * 2.208

= 1.8768 m/s

4.3.18. Calculation of Maximum Volumetric Flow rate

Q = m / ρ

For Top:

Q = 5911.239 / 0.5896

Q = 10025.84 m^3/hr or 2.784 m^3/s

For Bottom:

Q = 5911.239 / 2.4961

Q = 2368.18 or 0.6578 m^3/s

4.3.19. Net Area Required

Net Area = $\frac{Maximum\ Volumetric\ flowrate\ of\ vapor}{Design\ Velocity\ of\ Vapor}$

For Top:

Net Area = $\frac{2.784}{3.3575}$

Net Area = 0.8291 m^2

For Bottom:

Net Area = $\frac{0.6578}{1.8768}$

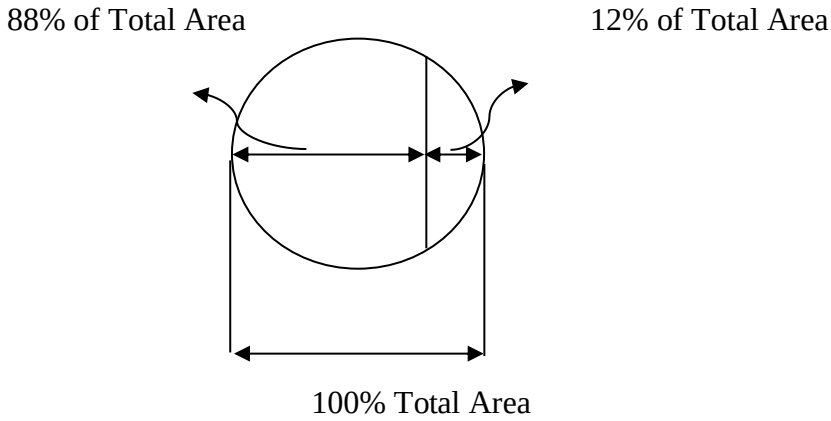
Net Area = 0.3504 m^2

4.3.20. Column Cross-Sectional Area

Taking Down comer Area as 12% of Total Area:

Down comer Area = 12% Net Area

So, Net Area will be (100 -12) % = 88%



Cross-Sectional Area of Top of Column:

$Area_{Cross\text{-}Sectional}$ = Net $Area_{Top}$ / 0.88

$Area_{Cross\text{-}Sectional}$ = 0.8291 / 0.88 = 0.942 m^2

Cross-Sectional Area of Bottom of Column:

$Area_{Cross\text{-}Sectional}$ = Net $Area_{Bottom}$ / 0.88

$Area_{Cross\text{-}Sectional}$ = 0.3504 / 0.88 = 0.3981 m^2

4.3.21. Diameter of the Column

Area = $\frac{\pi}{4}(Diameter)$^2

Hence,

Diameter = $\sqrt{\frac{4*Area}{\pi}}$

Diameter of the Top of the Column:

$Diameter_{Top}$ = $\sqrt{\frac{4*0.942}{\pi}}$

$Diameter_{Top}$ = 1.095m

Diameter of the Bottom of the Column:

Diameter = $\sqrt{\frac{4*0.3981}{\pi}}$

$Diameter_{Bottom}$ = 0.7119m

4.3.22. Finding the Liquid Flow Pattern

For liquid flow pattern, the maximum volumetric flowrate of liquid at the bottom is required:

$Q = m/\rho_L$ or $Q = L_m/\rho_L$

Where,

m or L_m is the liquid flowrate at the bottom of the column,

ρ_L is the liquid density at the bottom of the column.

Hence,

$Q = (\frac{9244.569}{665.59}) = 13.8\ m^3/hr$ or $0.00385\ m^3/s$

Using the Bottom diameter of the column ($Diameter_{Bottom}$ = 0.7119m) and the maximum Volumetric Flow rate of the liquid at the bottom of the column to find Liquid Flow Pattern from the Graph (Figure 17.35 in Book: Chemical Engineering design by Galvin Towler & Ray Sinnott)
(Sinnott, Galvin Towler and Ray 2002)

From the Figure 17.35 in book: Chemical Engineering design by Galvin Towler & Ray Sinnott (Sinnott, Galvin Towler and Ray 2002). The Bottom diameter is outside the region of Figure, but Single Pass plate can be used, as it can be seen from the figure.

4.3.23. Calculation of the Weeping velocity

At bottom of the Column:

Diameter = 1.095m

Column Area = A_c= $0.942m^2$

Down comer Area = A_d= 12% of Column Area = 0.12 * 0.942 = $0.1130m^2$

Active Area = $A_a = A_c – 2A_d$ = 0.942 - 2*(0.1130) = $0.716m^2$

Net Area = $A_n = A_c – A_d$ = 0.942 – 0.1130 = $0.829m^2$

Hole Area (as first Trial) = A_h = 10% of the Active Area

A_h = 0.10 * 0.716 = 0.0716m^2.

4.3.24. Calculating Weir Length

$(\frac{Ad}{Ac})$ * 100 = $\frac{0.1130}{0.942}$ *100 = 12

Locating L_w/D_c on the Figure 17.39 in book: Chemical Engineering design by Galvin Towler & Ray Sinnott, by using $(\frac{Ad}{Ac})$ * 100 and the curve:

(Sinnott, Galvin Towler and Ray 2002)

Hence, from graph:

$\frac{lw}{Dc}$ = 0.765

Where,

l_w is the Weir Length,

D_c is the Diameter of the Bottom.

l_w = 0.765 * D_{Top} = 0.765 * 1.094

l_w = 0.8369m

Maximum Liquid rate of Bottom = 9244.569 kg/hr or 2.5679 kg/s

Maximum height of the liquid Crest over the Weir = $Maximum_{how}$= 750*$(\frac{Lw}{\rho L * lw})$

Where,

L_w is the liquid flowrate in the bottom,

l_w is the Weir length,

ρ_L is the liquid density at the bottom.

Hence,

$Maximum_{how}$ = $(\frac{2.5679}{663.59*0.8369})$^(2/3) = 20.816mm liquid

Taking turn down ratio = 70% (Example 17.6 in Book: Chemical Engineering design by Galvin Towler & Ray Sinnott) (Sinnott, Galvin Towler and Ray 2002)

Minimum Flow rate at 70% Turndown ratio: 0.70 * 2.5679 = 1.7975 kg/s

$Minimum_{how} = (\frac{1.7975}{663.59*0.8369})$ ^ (2/3) = 16.41mm liquid

At minimum Flow rate: $h_w + h_{ow}$

Where,

h_w is the weir Height,

h_{ow} is the height of liquid crest over the weir.

Using Weir Height = 50 mm (recommended in section 17.13.8 in Book: Chemical Engineering design by Galvin Towler & Ray Sinnott)

(Sinnott, Galvin Towler and Ray 2002)

$h_w + h_{ow} = 50 + 16.41 = 66.41$ mm

Using the value ($h_w + h_{ow}$) and finding the value of K_2 from the figure 17.37 in Book: Chemical Engineering design by Galvin Towler & Ray Sinnott:

$K_2 = 30.5$

Weeping Velocity is calculated as:

The minimum design velocity is given by:

$u_h = \frac{(K2-0.90(25.4-dh))}{(\rho V)\text{^}0.5}$

where,

u_h is the minimum vapor velocity through the Holes (based on the Hole area),

d_h is the Hole diameter = 5mm (recommended fin section 17.13.10 in Book: Chemical Engineering design by Galvin Towler & Ray Sinnott),

K_2 is the constant dependent upon the depth of the clear liquid on the plate found by Figure 17.37 in above.

Hence,

$u_h = \frac{(30.5-0.90(25.4-5))}{(2.496)\text{^}0.5}$

$u_h = 7.842$ m/s

Actual minimum velocity = $\frac{minimum\ vapor\ rate}{Ah}$

= $\frac{0.7*\ 0.6578}{0.0716}$ = 6.4310 m/s

As the actual minimum velocity is less than the minimum vapor velocity, therefore it will cause weeping. Hence, take hole area as the 7% of the active area:

A_h = 0.07 (0.716) = 0.05012

New Actual minimum velocity = $\frac{0.7*0.6578}{0.05012}$ = 9.1871 m/s.

This minimum vapor velocity is well above the weep point.

4.3.25. Calculation of Plate Pressure drop

4.3.26. Dry plate drop

Maximum vapor velocity through the holes:

$u_{h(max)} = \frac{Maximum\ vapor\ flowrate\ in\ bottom}{Ah}$

$u_h = \frac{0.6578}{0.05012}$ = 13.124 m/s

Calculation of the Down comer Liquid Backup:

$h_b = (h_w + h_{ow}) + h_t + h_{dc}$

Where,

h_t is the total plate drop,

h_b is the downcomer backup, measured from the plate surface,

h_{dc} is head loss in the downcomer

Now,

h_w = 50 mm,

h_{ow}(max) = 20.816mm

$h_{dc} = 166(\frac{Lwd}{\rho L*Am})$

where,

L_{wd} is the liquid rate in the downcomer,

$A_m = A_{ap} = h_{ap} * l_w$

Where,

L_w is weir length

$h_{ap} = h_w – 10$ (taking the value = 10 from the range 5 to 10mm)

$h_{ap} = 50 – 10 = 40mm = 0.040m$

So, $A_m = A_{ap} = 0.040 * 0.8369 = 0.0334\ m^2$ (Since A_m is less than $A_d = 0.1130$, so use A_m in the relation to find h_{dc})

$h_{dc} = 166((\frac{2.5679}{663.59 * 0.0334})^2) = 2.228mm$

$h_t = h_d + h_r + (h_w + h_{ow})$

$h_d = 51(\frac{uhmax}{Co})^2(\frac{\rho V}{\rho L})$

For C_0 using $\frac{Ah}{Ap}$ or $\frac{Ah}{Aa}$ and plate thickness/hole diameter = 1 and finding C_0 from figure 17.42 in Book: Chemical Engineering design by Galvin Towler & Ray Sinnott: (Sinnott, Galvin Towler and Ray 2002)

Hence,

$\frac{Ah}{Aa} = \frac{0.05012}{0.716} = 0.0715$, so $C_0 = 0.82$

Hence,

$h_d = 51(\frac{13.124}{0.82})^2(\frac{2.496}{663.59}) = 49.138$ mm liquid

Residual head = $h_r = \frac{12500}{\rho L} = \frac{12500}{663.59} = 18.8$ mm liquid

4.3.27. Backup in downcomer

$h_b = (50 + 20.816 + (49.138 + 18.8 + (50 + 20.816)) + 2.228)$

$h_b = 211.798$ mm

As,

$h_b < \frac{1}{2}$(Plate spacing + weir height)

211.798 mm < 275 mm

Hence, this Plate Spacing is Acceptable.

4.3.28. Residence time

$h_r = \frac{Ad*hbc*\rho L}{Lwd}$

$h_r = \frac{0.1130*0.2117*663.59}{2.5679} = 6.18$ s

The residence time is greater than 3 seconds. Hence it is satisfactory.

4.3.29. Calculation of Entrainment

Percent Flooding: $\frac{un}{uf}$

Where,

U_n is the actual velocity based on the net area,

U_f is the flooding velocity.

Hence,

U_n = 0.6578/0.829 = 0.7934 m/s.

Percent Flooding = $\frac{0.7934}{2.208}$

Percent Flooding = 35.93 %

Using the value of F_{LV}=0.09577 and Percent flooding, we get the value of fractional entrainment from the Figure 17.36 in Book: Chemical Engineering design by Galvin Towler & Ray Sinnott:

Hence, Fractional Entrainment (Ψ) = 0.031 (it is below 0.1, which is satisfactory) (Sinnott, Galvin Towler and Ray 2002)

4.3.30. Calculation of the Number of Holes

Number of Holes = $\frac{Ah}{Ah'}$

Where,

A_h is the overall hole area = 0.0716 m^2

A_h' is the Area of a single Hole = 1.964 * $10^{-5} m^2$.

Number of Holes = $\frac{0.0716}{0.00001964}$ = 3645.62

4.3.31. Calculation of the Height of the Column

$H_c = (N_{Actual} – 1) *H_s + \Delta H$

Where,

$(N_{Actual} -1)$ is the Number of actual trays minus the reboiler,

H_s is the plate Spacing,

ΔH is the Top clearance + Bottom Clearance

Assuming,

Top Clearance = 1m

Bottom Clearance = 1m

Hence,

$H_c = (15 – 1) *0.5 + 2$

$H_c = 9$ m

Hence, the height of the column is 9 m.

4.4. Absorber Design

4.4.1. Theoretical Background

The process of gas absorption is the separation of a desired component from other components in the gaseous mixture, by using a suitable solvent. It is one of the major operations, which are named as the separation processes in chemical engineering.

The gas absorption process involves re-distribution of solute between gas phase and the liquid phase when the two phases come into close contact and achieve equilibrium condition.

4.4.2. Types of Absorption

Physical Absorption:

Diffusion is the working principle of the physical absorption, followed by physical equilibrium.

Chemical Absorption:

In this type of absorption as soon as a particular component comes in contact with the absorbing liquid a chemical reaction take place, thus reducing the concentration of component in liquid phase, which enhances the rate of diffusion.

Selection of Absorbent:

Choice of a suitable solvent completely depends upon the operating temperature and pressure in the equipment, and the level of the purity to be achieved.

Properties of an ideal Absorber:

- Solute is highly soluble.
- Less volatile to prevent loss of solvent.
- Non-reactive and stable.
- No or less corrosion
- Less viscous and highly diffusive.
- Low foaming proclivities
- less toxic and not flammable
- Available in the process (if possible)
- Less costly

4.4.3. Types of Absorbers

Mainly, there are two types of absorber column used in industries

- Tray Column
- Packed Column

Tray Column

These columns consist of trays with suitable tray spacing (range between 0.15m -1m). The height of the Distillation column depends upon the plate spacing. The plate spacing depends upon the diameter of the column and the operating conditions. A tray column mainly uses three types of trays:

1. **Sieve-plates** (the vapor passes through the holes and the liquid is retained on the plate, the liquid starts weeping through the holes at low flowrates, hence reducing the plate efficiency).
2. **Bubble-cap Plates** (the vapor passes through small pipes, which are called risers, covered with caps such that there is a slot between the riser wall and the edge of the cap which allow the vapors to flow. This design has a good turn down ratio even at low flowrates, high liquid-vapor Contact time).
3. **Valve-Plates** (Sieve Plates with movable flaps which lift according to flow of the vapors through the holes. They offer less contact time than the bubble-cap plates, pressure drop is high).

Packed Column:

The working of packed column is based on the continuous Liquid-Vapor contact throughout the packed bed. The Liquid flows from the to the bottom of the column and gas flows counter-currently in the direction of the liquid flow. The performance of a packed column mainly depends upon the good Liquid-Vapor distribution and contact in throughout the Packed bed. The Packed Columns mainly contains a liquid distributor, a hold down plate and the Packing support.

4.4.4. Absorber Selection

The choice of device is most often between a trayed and a packed column. The comparison of favorable conditions for both is given:

4.4.5. Selection of Tray Column:

- The material that is used in absorption that is corrosive and toxic so it may damage the packing, if we use metallic packing it because rusting.
- Plates should always be considered for large diameter columns, say greater than 0.6m, where plates would be easy to install and less expensive. So, our diameter is 1.4693m that is why we selected tray column.
- High temperature may cause distortion of the packings used, in case of packed column. As we are introducing the vapor at 90^0 C so it may harm the packing.
- The pressure drop per equilibrium stage (HETP) can be lower for packing than plates and packing should be considered for vacuum columns. As pressure is high plate column is selected.
- Plate columns are more suitable for handling non-foaming systems as the system is non-foaming therefore plate column is selected.

4.4.6. Given Data of Absorber

Flow rate of liquid (water) Inn from top = 2438.773006 kg/h

Flow rate of liquid (water + maleic anhydride) out from bottom = 5568.529006 kg/h

Vapor flow rate out at top (Mixture of air) = 80321.79 kg/h

Vapor flow rate inn from bottom (Mixture of air, water, and Maleic Anhydride) = 83451.55kg/h

Vapor density= 38.1355 kg/m^3

Liquid density= 995.65 kg/m^3

Surface tension of liquid= 71.03 J/m^2 or dyn/cm^2

tray spacing= 0.15 - 1 meter (Reference, *Chemical process equipment, chapter Rule of Thumb 1 page XV Point number 13)*

Downcomer area= 7-12% of column area (Reference *Chemical Engineering Design Principles page 867*)

For primarily design we take 12%

Whole diameters are smaller than for vapor-liquid contacting, being 3–8 mm.

For primarily design mostly used hole size = 5mm

Residence time is at least 3 seconds is recommended (Reference *Chemical Engineering Design Principles page 877*)

4.4.7. Diameter of Column

$$\text{Area of column} = A_c = \pi D_c^2/4$$

Diameter of the column = $D_c = \sqrt{\frac{4\,Ac}{\pi}}$

Active Area of the Column = $A_a = \frac{\text{Volumetric Flow rate of gas (Q)}}{\textit{Design Velocity}\ (\text{Vd})}$

V_D is 60-80% of flooding velocity

Flooding velocity = $V_F = C_{SB}\,(\sigma/20)^{0.2}\sqrt{\frac{\rho\ \text{liq} - \rho\ \text{vap}}{\rho\ \text{vap}}}$ (DON W. GREEN n.d.)

σ = surface tension

$\boldsymbol{\rho_{liq}}$ = density of liquid

$\boldsymbol{\rho_{vap}}$= density of vapor

C_{sb} = Souder's brown flooding Constant

For Souder's brown flooding Constant we required Liquid Vapor Factor (F_{LV})

$F_{LV} = \frac{L}{V}\sqrt{\frac{\rho\ \text{vap}}{\rho\text{liq}}}$ (DON W. GREEN n.d.)

We need to calculate the F_{lv} for Souder's brown flooding Constant to be used in eq

F_{LV} at bottom of column:

$F_{LV} = \frac{2438.773}{83451.55}\sqrt{\frac{38.1355}{995.65}}$

F_{LV} = 0.010001

Souder's brown flooding Constant can be find by the graph given in *Perry Handbook Chapter 14 page 37*

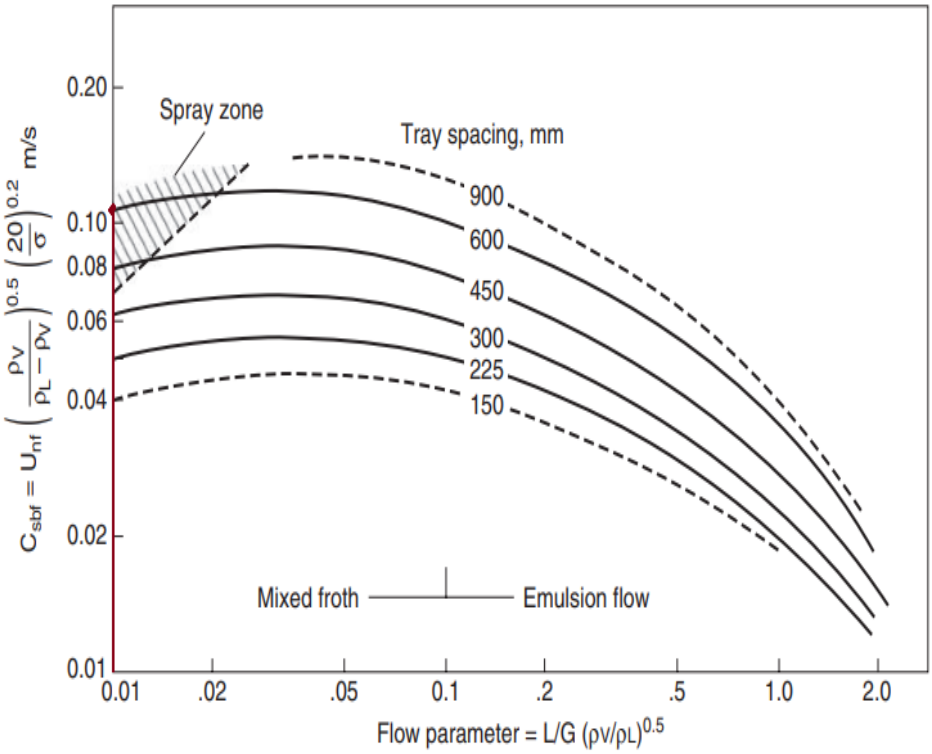


Figure 4.8: Graph for Souder's brown flooding constant

The value of C_{SB} from graph given above is 0.10 m/s as we take the tray spacing is 0.6 m (600mm)

By putting the value of C_{SB} in equation (3) we can calculate flooding velocity

$$V_F = 0.10\ (71.03\ /20)^{0.2} \sqrt{\frac{995.65-38.1355}{38.1355}}$$

V_F = 0.64564 m/s

We take the design velocity 80% of flooding velocity

V_D = 0.64564 * 0.8

V_D = 0.517 m/s

Volumetric Flow rate of Gas = $\frac{\text{mass flow rate } (\dot{m})}{\text{density of vapor } (\boldsymbol{\rho}\ \mathbf{vap})}$

$$Q = \frac{83451.55}{38.1355*3600}$$

Q = 0.60786 m^3/s

By putting the value of design velocity and volumetric flow rate in equation (2) we will get the area of the column

$A_n = \frac{0.60786\ m3/s}{0.3934\ m/s}$

$A_n = 1.5451m^2$

Assuming downcomer occupies 12% of across sectional area so,

$Ac = A_n + A_D$

$A_C = A_n + 0.12Ac$

$A_C = 1.5451 / 0.88$

$A_C = 1.75585\ m^2$

Diameter can be determined by putting the value in equation (1)

Diameter of the column = $D_c = \sqrt{\frac{4*1.75585\ m2}{\pi}}$

$D_c = 1.4956m$

We will used tray tower as the Diameter of the column is greater than 0.6 meter

Number of theoretical stages

Absorption factor: $A_i = \frac{L}{KV} = 1.5$

Generally, the design that minimizes the combination of operating costs (i.e., the consumption of separating agent) and the capital costs (i.e., the number of plates) occurs when KV/L is chosen to be between 1.5 and 3 for an absorber.

Using Kremser equation for calculation of number of stages: $\frac{(A_i^{N+1} - A_i)}{(A_i^{N+1} - 1)} = \varepsilon$

Simplifiying above equation: $N + 1\log(A_i) = \log\frac{(A_i - \varepsilon)}{(1 - \varepsilon)}$

Solute fraction absorbed = **ε** = 0.988

$N = \log(\frac{1.5-0.988}{1-0.988}) * (\frac{1}{\log(1.5)}) - 1$

N = 8.25

N = 9

Number of actual stages = $\frac{Theoretical\ stages}{Efficiency}$ (*eq 17.32*)

(Sinnott, Galvin Towler and Ray 2002)

The efficiencies of the columns lies between 30% to 80%, but usually selected as 70%.

(*Reference Chemical Engineering Design - Principles, Practice and Economics of Plant and Process Design 2nd ed - G. Towler, R. Sinnott, Chapter 17 Pg 844)*

Efficiency = 70%

$N_{actual} = \frac{9}{0.70}$

N_{actual} = 12.8571

N_{actual} = 13

4.4.8. Height of Column

Height of column = 13 * 0.6

H_c = 7.8m

4.4.9. Plate Design

Table 4.5: Comparison of three Types of trays

Parameter	*Bubble-cap tray*	*Sieve tray*	*Valve tray*
Capacity	Moderate	High	High to very high
Efficiency	Moderate	High	High
Entrainment	High (about three times that of a sieve tray)	Moderate	Moderate
Pressure drop	High	Moderate	Moderate
Turndown	Excellent (can operate at a very low capacity)	About 2 (not suitable to operate at variable loads)	4-5
Fouling tendency	High, tends to collect solids	Low	Low to moderate
Cost	High (about 2–3 times that of a sieve tray)	Low	About 20% more than sieve trays
Applications	Rarely used in new columns, may be used if low flow rates are anticipated	Most applications if turndown is not important	Preferred if a high turndown is anticipated
Share of market	About 5%	25%	About 70%

Selection of Sieve Tray

- It has high capacity and high efficiency than valve and bubble cap tray.
- It has moderate pressure drop and moderate entrainment.
- It has low cost and low fouling tendency.

Column Diameter = D_c= 1.4956m

Column Area = A_C = 1.75585 m^2

Downcomer Area =A_d = 0.12A_c = 0.12*1.75585m^2

Downcomer Area = A_d= 0.2107 m^2

Active Area = A_a = A_C – 2A_d = [1.75585 – 2(0.2107)] m2

Active Area = A_a = 1.33445 m^2

Hole size or diameter typically 5mm hole size is used.

The weir heights will normally be between 40 mm to 90 mm (1.5 to 3.5 in.); 40 to 50 mm is recommended. We are taking 50mm.

Typical plate thicknesses used are: 5 mm (3/16 in.) for carbon steel.

(Sinnott, Galvin Towler and Ray 2002)

(Reference Chemical Engineering Design - Principles, Practice and Economics of Plant and Process Design 2nd ed - G. Towler, R. Sinnott, Chapter 17 Pg 871,872)

D_h = 0.005

Weir Height = H_w = 0.05 m

Plate thickness = 0.005m

For an equilateral triangular pitch

P_h = 2.5 to 4.0 times of hole diameter

P_h = 2.5 * 0.005

P_h = 0.0125m

Weir Length

Weir length, L_w=77% x Dc

Weir length, L_w = 0.77 x 1.4956 m

L_w = 1.151612 m

4.4.10. Number of holes per plate

Holes area, A_h = 10% of active area

A_h = 0.1* 1.33445 m^2

A_h = 0.133445 m^2

Area of one hole = $\frac{\pi}{4}(D_h^2)$

$A_h = \pi*(0.005)^2/4$

A_h = 0.0000196 m^2 = 1.96 X 10^{-5}

Number of holes = Area of holes/ Area of one hole

Number of holes = $\frac{0.133445}{0.0000196}$

Numbers of hole = 6808.41 = 6808

4.4.11. Pressure Drop Calculation

Total pressure drop

$$H_t = h_d + (h_w + h_{ow}) + h_r$$

Now we will find all these Pressure drops one by one.

h_w and h_{ow}:

h_w = weir height, that is 15% of plate spacing

h_w = 0.15 * 0.6

h_w = 0.09

$h_{ow} = 750\ [L_w / \boldsymbol{\rho}liq * l_w]^{2/3}$

lw = weir length

Lw = liquid flowrate

$\boldsymbol{\rho}$liq = 985.7kg /m^3

Weir length = 0.77 * Diameter of column

= 0.77 *1.4956m

= 1.151612m

h_{ow} = 750 [0.67744 / 985.7*1.151612]

= 0.02476 m

$h_w + h_{ow}$ = 0.09 + 0.02476 = 0.11476m = 114.76mm

$h_d = 51\ [U_h / C_o]\ \boldsymbol{\rho}\mathrm{vap}/\boldsymbol{\rho}\mathrm{liq}$ (17.52)

h_d = dry plate pressure drop

U_h = Minimum design vapor velocity through hole.

C_o = Orifice plate thickness. It is function of plate thickness, hole diameter, hole to perforated ratio that will be obtain from the graph. (fig 17.42)

$$U_h = \frac{K2 - 0.90\ (25.4 - Dh)}{\sqrt{\boldsymbol{\rho}\mathrm{vap}}}$$

K_2 = Constant, dependent on depth of clear liquid on plate.

D_h = hole diameter, mostly used 5mm for initial design.

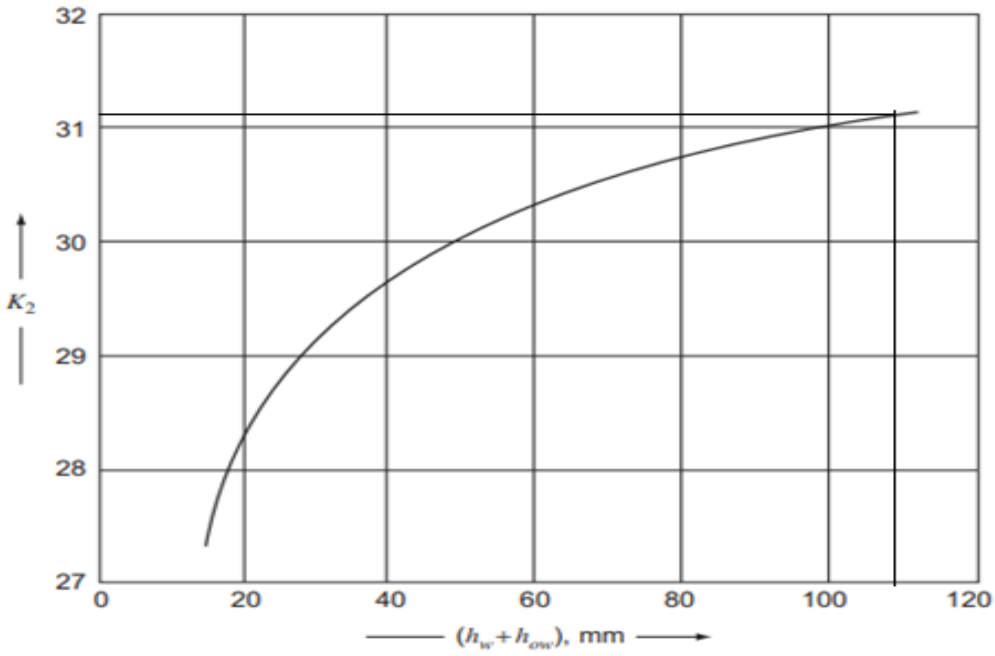


Figure 4.9: Graph for value of K_2

From above graph the value of K_2 is 31.2.

By putting this value of K_2 in equation (6)

$$U_h = \frac{31.2 - 0.90\,(25.4 - 0.05)}{\sqrt{38.1355}}$$

U_h = 1.3578m/s

C_o will be obtained from graph fig (12.42, Chemical Engineering Design - Principles, Practice and Economics of Plant and Process Design (Sinnott, Galvin Towler and Ray 2002)

$A_h / A_p = 0.9\,[D_h / L_p]^2$

$= 0.9\,[0.05 / 0.125]^2$

= 0.144

Perforated area = A_h / A_p *100

= 0.144*100 = 14.4

Plate thickness / hole diameter = 5mm /5mm = 1

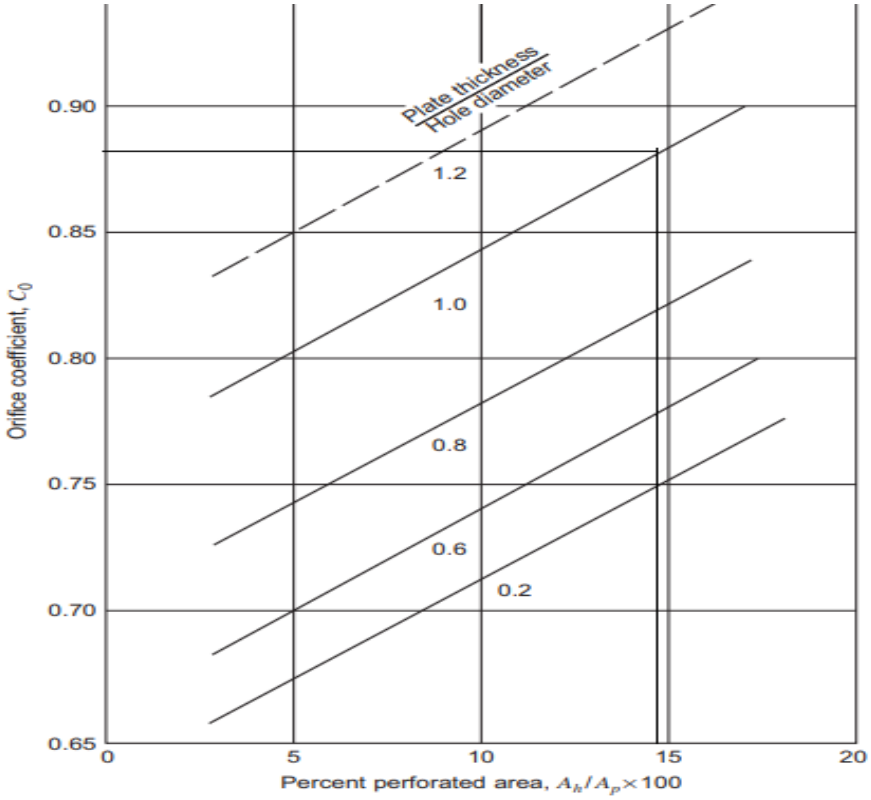


Figure 4.10: Graph for value of C_0

The value of C_o is 0.88

By putting the vales in equation (5) we will get the pressure drop through dry plate

$h_d = 51\ [1.3578 / 0.88]\ \frac{38.1355}{985.7}$

$h_d = 0.3044533m$

$h_r = 12.5 * 10^3 / \rho L$

$h_r = 12.5 * 10^3 / 985.7$

$h_r = 0.01268$

We will obtain Pressure drop by putting values in equation (4)

$h_{total} = 0.01268 + 0.3044533 + 0.11476m$

$h_{total} = 0.4318933m$

$h_{total} = 431.8933mm$

4.4.12. Chemical Absorption Design:

Now using below equation to calculate C_g:

$$PV = nRT$$

Rearranging the above ideal gas equation

$$n/V = P/RT$$

$$C_g = P/RT$$ (levenspiel n.d.)

Where,

P= 1 bar

R= 0.08314 L.bar/mol.K

T= 90+273= 363 K

After calculations,

$$C_g = 0.03313 \text{ mol/L or mol/dm}^3$$

Molar flow rate of solvent (at inlet) $=F_L$= 135.48 kmol/h

Molar flow rate of gas (at inlet) $=F_g$= 300 kmol/h

Molar flow of bottom = 21.013 kmol/h

Area = A= 1.75585m^2

F_L/A= 77.16 kmol/h.m^2

F_g/A= 170.85 kmol/h.m^2

$$P_{A1} - P_{A2} = ((F_L/A).\pi / (F_g/A).C_T) * (C_{A3}-C_{A1})$$ (levenspiel n.d.)

Pressure at outlet of gas $=P_{A1}$ = 0.95755 bar

Pressure at inlet of gas $=P_{A2}$ = 1 bar

C_{A3}=?

Initial conc. $=C_{A1}= 0$

C_{Ttop} = Conc. of solvent at Top

$$\mathbf{C_T =(F_L/L_W)* \rho_L}$$ (levenspiel n.d.)

$C_T = 55.31$ mol/m^3

Now put all the values in above equation, to calculate C_{A3}:

$C_{A3}= 34293.55$ mol/m^3

K_g= mass transfer coefficient of gas

K_L= mass transfer coefficient of liquid

$$\mathbf{K_g=F_g/P_{A3}*A}$$ (levenspiel n.d.)

K_g= 526.766 mol/h. bar

$$\mathbf{K_L=F_L/A* \Delta C_A}$$ (levenspiel n.d.)

ΔC_A = ?

For calculating ΔC_A

$C_{Tbottom}$ = molar flow of bottom/mass flow of bottom * density

C_T = 21.013/2438.773 * 995.65

C_T = 8.5787 mol/m^3

Now for $C_{Lbottom}$:

$C_{Lbottom}$= 21.013/2438.773 * 995.65 = 8.5787

Mole fraction of solvent in bottom = 0.0714

So,

K_L= 1381.90/0.490*(12,453-11,712.04)

$K_L = 0.0126$

Calculating Henry's constant:

$h_A = 0.159*2.5e5/34293.55 = 1.15$

$\mathbf{-r_m = h_A * C_B + P_A/(1/K_{AG}*a + 1/K_{AL}*a)}$ (levenspiel n.d.)

= 1.115*34293.55+1/(1/3.35*100)+(1.15/0.0126*100)

= 9.084

$\mathbf{V_r = F_g/P * 1/-r_m*(P_{A2}-P_{A1})}$ (levenspiel n.d.)

= 4.108/1 * 1/94.084*(0.04245)

= 9.6758

H= height = $\mathbf{V_r/A}$

= 13.6458/1.75585

= 7.7144m

4.5. Flash Vessel

4.5.1. Theoretical Background

Flash steam is created when high pressure condensate is released to a lower pressure. The word 'flash' depicts the way it is shaped. At air pressure, water bubbles at 100 Deg. C. Inside steam channeling or any compressed vessel, steam is by and large used at a pressure over the environmental pressure. At the point when this steam loses its heat either by moving it to the item being heated or by radiation misfortune to the climate, condensate is framed. This condensate which is shaped, is additionally at same pressure and temperature as that of steam (Bryant 1984). At the point when this compressed condensate is presented to climatic pressure, it has energy beyond what it can contain at air pressure. This abundance energy is utilized to change over a piece of this condensate into steam. This peculiarity is called as flashing and the steam so created is alluded to as flash steam. For example, think about a cycle where steam is utilized at a pressure of 6 bar. At this pressure, the enthalpy of steam would be 659.33 Kcal/Kg. In the wake of losing its inactive heat to the item being heated, condensate will be shaped at same pressure with an enthalpy of 160.53 Kcal/Kg. Allow us to expect, after the snare, this condensate is released to a pressure of 1 bar. According to the steam tables, water in fluid structure can contain 99.76 Kcal/Kg of enthalpy. As it very well may be seen, there is 60.77 Kcal/Kg of overabundance energy. This energy will change over a piece of the fluid condensate into fume (Sarco 2014).

Flash steam is low-pressure steam made when heated water is set free from a high pressure to a lower pressure inside a steam framework. For instance, it tends to be delivered through a steam trap or from heater blowdown. Flash steam is now and again considered a side-effect with little worth contrasted with live steam directly from the kettle. This can be a costly error. Steam, but created, can be an effective heat transporter. As well as being inefficient, the venting of flash steam is unattractive and can be both hazardous to individuals and harming to a structure's texture. A long way from being waste, flash steam is a significant wellspring of heat.

A flash steam recuperation framework discharges flash steam into a vessel, as opposed to air. This implies it very well may be reintegrated into the steam framework or used to take care of low-pressure applications, for example, space heating. Condensate, which is now flashing as it leaves steam traps on high-pressure units, enters the flash

vessel generally mostly up the steam and condensate separate to the top and lower part of the vessel individually.

'Flash steam' is let out of hot condensate when its pressure is diminished. Indeed, even water at a surrounding room temperature of 20°C would bubble on the off chance that its pressure was brought down sufficiently far. It could be actually significant that water at 170°C will bubble at any pressure beneath 6.9 bar g. The steam delivered by the flashing system is all around as valuable as steam set free from a steam heater (Sarco 2014).

A condensate flash separator vessel is the significant part of a flash recuperation framework. Many plants use them to recuperate the energy that is in their condensate. The siphons drive the higher strain condensate into a lower pressure flash tank to achieve this energy recuperation. Then, at that point, at this lower pressure, a level of the condensate flashes (Bryant 1984). This flash steam is an asset. For example, you can involve it for low-pressure cycles or warming. Then, there is the choice to rehash the cycle briefly stage flash, at an even lower pressure. Subsequently, you have further developed effectiveness and cost investment funds. The cycle begins with the utilization of an inlet nozzle directed tangentially. Accordingly, the condensate is placed in a turning movement as it enters the lower pressure tank. This turning movement presses the fluid against the external dividers of the tank through diffusive power. This outcomes in a low-pressure vortex in the vessel, which the steam flashes into. Then, at that point, an outlet spout at the focal point of the vortex conveys the flash steam up and out of the tank. Ultimately, the leftover condensate channels out the lower part of the tank. Utilizing this turning movement permits us to flash the steam at a high speed.

4.5.2. Applications of Flash Steam

Continuous blowdown of boiler water is important to control the degree of TDS (Total Dissolved Solids) inside the boiler. Continuous blowdown fits the recuperation of the heat content of the blowdown water and can empower significant investment funds to be made. Boiler blowdown contains gigantic amounts of heat, which can without much of a stretch be recuperated as flash steam. After it goes through the blowdown control valve, the lower pressure water streams to a flash vessel. Now, the flash steam is liberated from pollution and is isolated from the condensate and can be utilized to heat

the boiler feed tanks. The lingering condensate depleting from the flash vessel can be gone through a plate heat exchanger to recover however much heat as could be expected before it is unloaded to squander. Up to 80% of the total heat contained in boiler continuous blowdown can be recovered thusly (Sarco 2014).

4.5.3. Design of Flash Steam Recovery Vessels

Before beginning the procedure of designing of flash steam vessels, we need to understand the methods of controlling flash steam pressure and factors on the sizing of steam vessels is dependent on.

How to control Flash Steam Pressure?

Sometimes, flash pressure will observe its own level and nothing more should be finished. At the point when organic market is generally in-sync, and especially on the off chance that the low-pressure steam is utilized on a similar gear delivering the high-pressure condensate, it is just necessary to pipe the flash steam to the low-pressure plant with no other control (Bryant 1984).

4.5.4. Sizing of Flash Steam Vessel

To size a flash vessel, the accompanying data is required:

- The steam pressure before the steam trap(s) providing the vessel.
- The total condensate flowrate into the flash vessel.
- The flash steam pressure in the flash vessel.

Joining this with estimating graph assists with deciding the size of vessel (Bryant 1984).

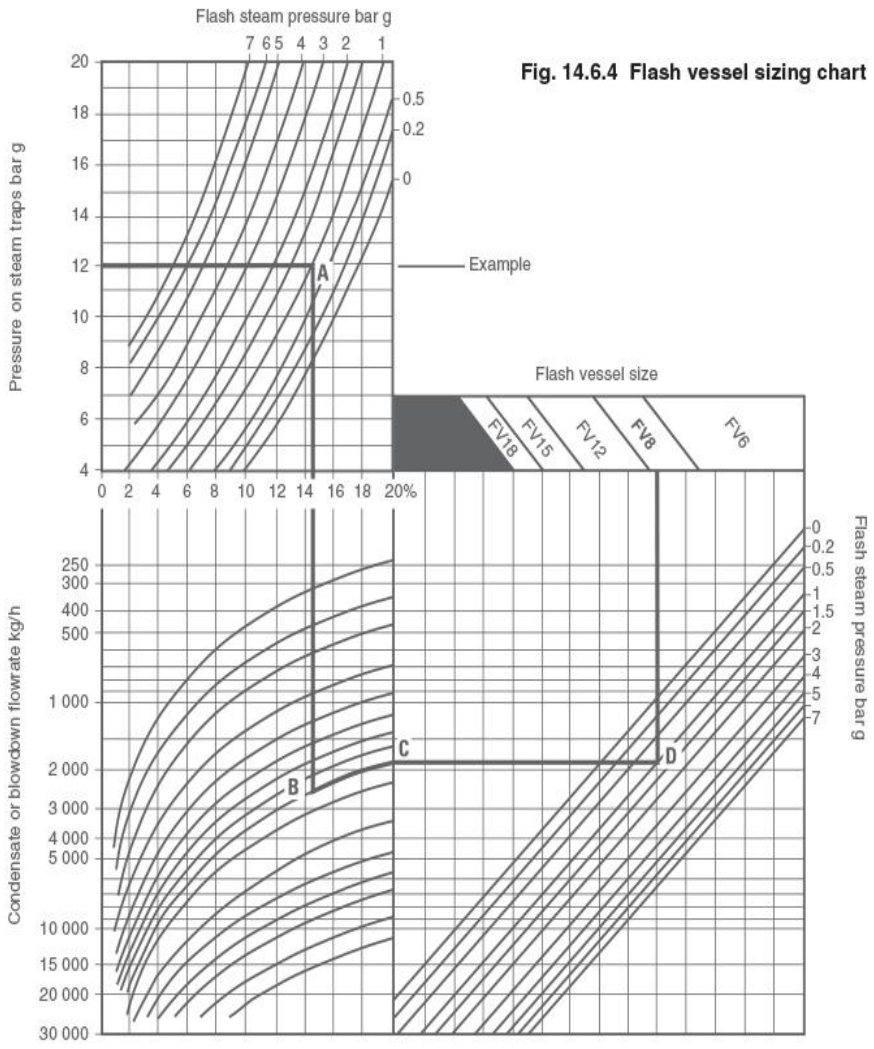


Figure 4.11: Flash Vessel sizing Chart

4.4.5. Utilization of Maximum Efficiency of Flash Steam

Assuming full use is to be made of flash steam, a few essential prerequisites should be fulfilled:

- It is fundamental to have a constant inventory of adequate condensate from applications working at higher pressures, to guarantee that enough flash steam can be delivered for monetary recuperation.
- Care should be taken while endeavoring flash steam recuperation with condensate from temperature-controlled gear. At not exactly full load, the steam space pressure will be brought down by the end activity of the steam control valve. Assuming the steam pressure in the hardware approaches or falls underneath the predetermined flash steam pressure, the general measure of flash

steam framed will be minor, and one should address whether recuperation is beneficial in this case.

- It is vital that there is an interest for low pressure flash steam that either rises to or surpasses the flash steam being created. Any shortfall of flash steam can be made up by live steam from a pressure lessening valve. Assuming that the stockpile of flash steam surpasses its interest, excess pressure will be made in the flash steam appropriation framework, which will then, at that point, must be vented to squander through a surplus valve.
- It is desirable over really utilize the flash steam near the high-pressure condensate source. Moderately huge breadth pipes are utilized for low pressure steam, to lessen pressure misfortune and speed, which can mean expensive establishment assuming the flash steam must be funneled any distance (Sarco 2014).

4.5.6. Given Parameters

The given parameters for Flash Vessel Design are:

Inlet Temperature: 65^0C

Outlet Temperature: 46.1^0C

Inlet Pressure: 2.1 Bar

Outlet Pressure: 1 Bar

4.5.7. Diameter of the Flash Vessel

Density of the vapor in the Top (N_2, O_2, CO_2, C_4H_{10}, H_2O, MAN): 1.057 kg/m^3.

Molecular Weight of the vapor in the Top = (0.004763*98.06) + (0.0327*18) + (0.02227*44) + (0.15058*32) + (0.7843+28) + (0.0049*58.12)

= 29.1197 kg/kmol.

Density of the liquid in the bottom (H_2O, MAN): 119.66 kg/m^3.

Average Molecular Weight of the Liquid in the Bottom = 91.65 kg/mol.

Vapor Flowrate = 86642.494 kg/hr.

We need the diameter of the flash vessel

Diameter = $\left(\frac{4*Area}{\pi}\right)^{\frac{1}{2}}$

For diameter firstly we need to calculate the area of the vessel

Area = $\frac{volumetric\ flowrate\ of\ Vapor}{design\ velocity\ *3600}$

For area we are required permitted velocity of fluid entering in vessel

Liquid vapor factor is required for calculating permitted velocity

$F_{LV} = \frac{MWliq}{MWvap}\sqrt{\frac{\rho v}{\rho L}}$ (Perry Handbook eq 14-89)

Putting values:

$F_{LV} = \frac{91.65}{29.1197}\sqrt{\frac{1.057}{119.66}}$

$F_{LV} = 0.027$

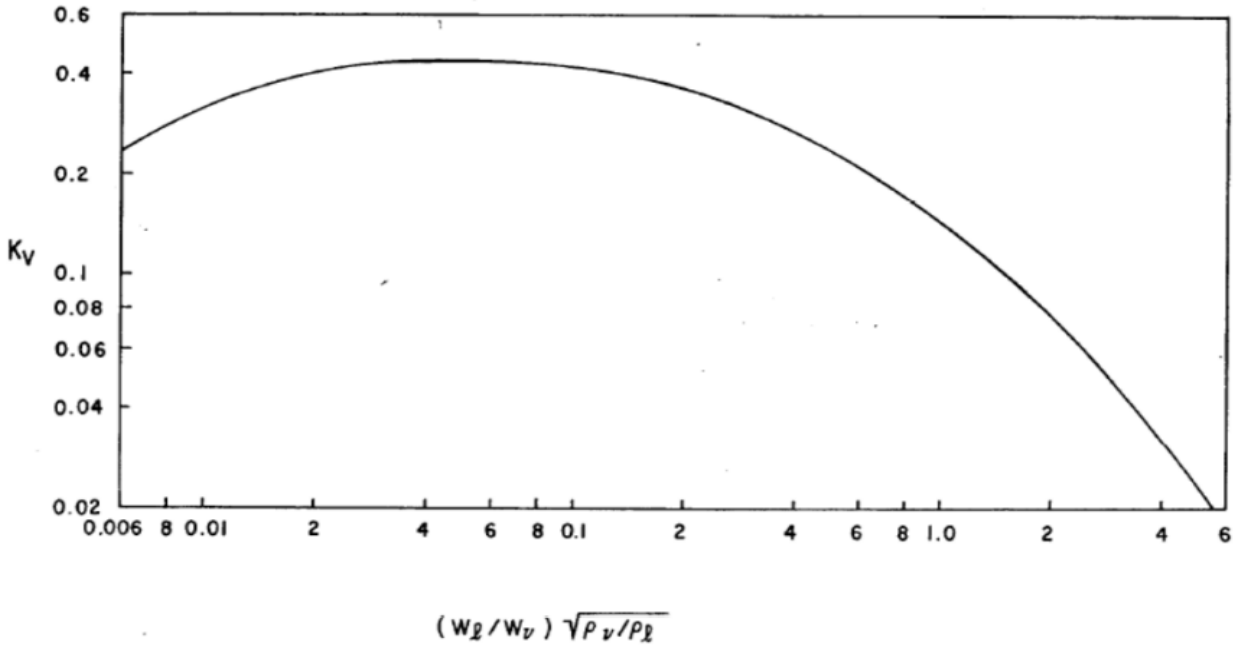


Figure 4.12: Graph for value of K

K = 0.4

For Permitted Velocity:

$\mu_{permitted} = K\left(\frac{\rho L-\rho V}{\rho V}\right)^{\frac{1}{2}}$

(DON W. GREEN n.d.)

$\mu_{permitted} = 0.4\left(\frac{119.66-1.057}{1.057}\right)^{\frac{1}{2}}$

$\mu_{permitted}$ = 5.48 m/s

Design velocity is 60% - 80% permitted velocity so the design velocity will be

design velocity = 5.48 * 0.8

design velocity = 4.384 m/s

volumetric flowrate = $\frac{Mass\ Flowrate\ feed}{\rho V*3600}$

volumetric flowrate = $\frac{86642.494}{1.057*3600}$

volumetric flowrate = 22.7694 m^3/s.

Area of the vessel = Volumetric flowrate / design velocity

Area of the vessel = 22.7694 / 4.384 = 5.194m^2

Diameter = $\left(\frac{4*Area}{\pi}\right)^{\frac{1}{2}}$

Diameter = $\left(\frac{4*5.194}{\pi}\right)^{\frac{1}{2}}$

Diameter = 2.5715 m

4.5.8. Length of Flash Vessel

Length = 3*D

Length = 3*2.5715

Length = 7.715 m.

4.5.9. Volume of the Flash Vessel

Volume = Length*Area

= (7.715 *5.194)

= 40.0700 m^3

4.5.10. Thickness of flash vessel

Internal pressure = 2.1 bar

Diameter of flash vessel = 2.5715m

Vertical stress thickness of the vessel can be calculated by

$$t = \frac{PD}{2ES - 1.2P}$$ (R K Sinnott 2005)

For longitudinal thickness we will use below relation

$$t = \frac{PD}{4ES + 0.8P}$$ (R K Sinnott 2005)

where;

P = internal pressure

D = diameter of the vessel

E = maximum allowable stress

S = welding efficiency of butt joint

As we know we have diameter and pressure, but we have to find maximum allowable stress(E) and welding efficiency of butt joint(S).

Some Maximum Allowable Stresses

Under ASME BPV Code Sec. VIII D.1, Taken From Sec. II Part D

Material	Grade	Min Tensile strength (ksi)	Min Yield strength (ksi)	Maximum temperature (°F)	Maximum allowable stress at temperature °F (ksi = 1000 psi)				
					100	300	500	700	900
Carbon steel	A285 Gr A	45	24	900	12.9	12.9	12.9	11.5	5.9
Killed carbon Steel	A515 Gr 60	60	32	1000	17.1	17.1	17.1	14.3	5.9
Low alloy steel 1 ¼ Cr, ½ Mo, Si	A387 Gr 22	60	30	1200	17.1	16.6	16.6	16.6	13.6
Stainless steel 13 Cr	410	65	30	1200	18.6	17.8	17.2	16.2	12.3
Stainless steel 18 Cr, 8 Ni	304	75	30	1500	20.0	15.0	12.9	11.7	10.8
Stainless steel 18 Cr, 10 Ni, Cb	347	75	30	1500	20.0	17.1	15.0	13.8	13.4
Stainless steel 18 Cr, 10 Ni, Ti	321	75	30	1500	20.0	16.5	14.3	13.0	12.3
Stainless steel 16 Cr, 12 Ni, 2 Mo	316	75	30	1500	20.0	15.6	13.3	12.1	11.5

Figure 4.13: Screenshot of Maximum allowable stresses for some materials

$T = 65^0C = 149^0F$

From above given graph

By interpolating between 100ksi and 300 ksi and converting ksi into psi we got the value

$E = 6.8*10^6$

For welding joint efficiency (S) we will choose double welded butt joint with fully radiographed. As it has high strength and high capacity to bear the temperature pressure conditions.

Sr. No.	Type of Welded Joint	Sketch	Weld Joint Efficiency, η		
			Fully Radio - graphed	Spot Radio - graphed	Not Radio - graphed
1.	Double Welded Butt Joint with Full Penetration		1.0	0.85	0.7
2.	Single Welded Butt Joint with Backing Strip		0.9	0.8	0.65
3.	Single Welded Butt Joint without Backing Strip.		–	–	0.60

Figure 4.14: Screenshot for Welding joint efficiency

Welding joint efficiency (S) = 1

Vertical thickness:

By putting the value in equation (14.13) we will get the vertical thickness.

$$t = \frac{210000 * 2.5715}{2*1*6800000 - 1.2(210000)} = 40.45\text{mm}$$

Longitudinal stress:

By putting the value in equation (14.14) we will get the longitudinal thickness.

$$t = \frac{210000*2.5715}{4*1*6800000 + 0.8(210000)} = 19.73 \text{ mm}$$

Chapter 5

Process Instrumentation and Control

For the process, to run efficiently and give maximum output, it is necessary for the process to be carried out at its specific process conditions. The efficiency of the process is maintained at its maximum by controlling the process at its defined process conditions. To control the process at its process conditions, it is required to apply Process Instrumentation and Control to the plant equipment.

Process Instrumentation basically involves the selection and implementation of the relevant and most suitable instrument to the equipment, which measures the variable to be controlled (Controlled variable).

Process Control involves the implementation of the Process Control Loops to the equipment, which consists of an Instrument, a Controller, and a Final Control Element. The measured value by the instrument is then transformed into an information signal (electric or pneumatic) which is then transmitted to the Controller which compares the signal with its set point and then transmits the signal to the Final Control Element (a Valve).

Process Instrumentation and Control on each equipment is necessary to control and operate the whole process on Optimum conditions. The Representation of the Process instrumentation and control on the Process Flow Diagram is represented in the form of a Piping and Instrumentation Diagram.

5.1. Process Instrumentation

Table 5.1: Process Instrumentation

Sr. #	Equipment	Temperature Sensor	Pressure Sensor	Level Sensor	Flow Sensor
1.	**P-01**		Differential Pressure Cell		
2.	**Comp-01**		Differential Pressure Cell		
3.	**Reboiler-01**	Thermocouple (T Type)			
4.	**M-01**				Coriolis Meter
5.	**HE-01**	Thermocouple (R or S Type)			
6.	**HE-02**	Thermocouple (R or S Type)			
7.	**P-02**		Differential Pressure Cell		
8.	**R-01**	Thermocouple (R or S Type)			
9.	**HE-03**	Thermocouple (R or S Type)			
10.	**HE-04**	Thermocouple (R or S Type)			
11.	**V-01**		Differential Pressure Cell		
12.	**HE-05**	Thermocouple (R or S Type)			
13.	**C-01**	Thermocouple (R 0r S Type)	Differential Pressure Cell	Guided Wave Radar	
14.	**M-02**				Coriolis Meter
15.	**HE-06**	Thermocouple (R or S Type)			
16.	**C-02**		Differential Pressure Cell	Guided Wave Radar	
17.	**HE-07**	Thermocouple (R or S Type)			
18.	**V-02**				Coriolis Meter
19.	**Reboiler-02**	Thermocouple (R or S Type)			Coriolis Meter

5.2. Process Control

The types of the Control systems used in this process flow diagram are discussed as:

5.2.1. Feed-Backward Control System

In this Control configuration, there are

- One Controlled Variable.
- One manipulated Variable.
- A final control element.

5.2.2. Auctioneering Control Configuration

In this control configuration, there are,

- One controlled variable, to be measured from every point along the equipment.
- Only one manipulated variable.
- A Final Control Element.

5.2.3. Ratio Control system

In this control configuration, there are

- Two controlled variables.
- A ratio controller.
- One manipulated variable.
- A final Control Element.

The process configuration of the various designed equipment are as follows:

5.2.4. Configuration for the Feed Salt Heat Exchanger

As per the requirement, to maintain the Outlet Temperature (420^0C) of the stream from the heat exchanger, the inlet flowrate of the stream is manipulated. Feed backward configuration is applied, as to control the output variable (Temperature) by manipulating the input variable (Flowrate). Hence, in this case,

- Control Objective is to Control the Temperature of the Outlet Stream.
- Controlled Variable is Temperature of the Outlet stream i.e., 420^0C.
- Manipulated Variable is Flowrate of the Inlet Stream.
- The Control Action is, if the Temperature rises, the flowrate decreases to maintain the temperature at 420^0C and vice versa.

- Simple Feed Backward Control system is used for this heat exchanger.

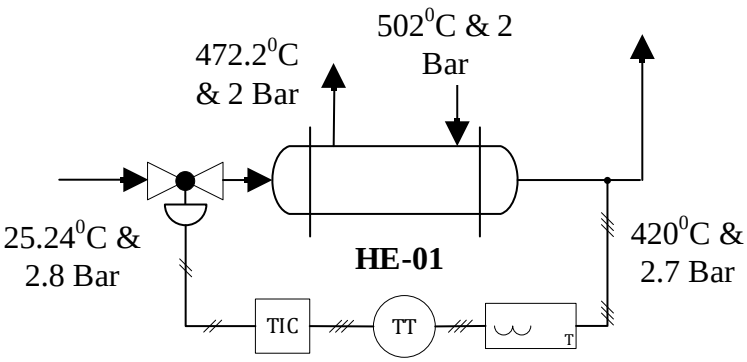


Figure 5.1: Control Loop on Feed Salt Heat Exchanger

5.2.5. Process Control Configuration for the Reactor

The optimum conditions for the desired reaction to be carried out is at 420^{0}C, while the reactor pressure must be controlled in the desired limits to prevent pressure buildup which may result in reactor explosion. So, it is necessary for the reactor conditions to be maintained at 420^{0}C (Feed Backward control system). The temperature sensed is transmitted to Temperature Controller which then transmits the information signal to the Final Control Element (the Valve of the Molten Salt Stream). The Pressure Differential Cell maintains the pressure by manipulating the Final Control Element (Pressure relief valve) by passing the information through the Pressure Controller. In this case,

For Temperature Control:

- Control Objective is to maintain the Reactor Temperature at 420^{0}C.
- Controlled Variable is Temperature of the Reactor.
- Manipulated Variable is the Valve of the Molten Salt Stream.
- Control Action is, when temperature of the reactor rises, the flowrate of the molten salt increases to maintain the temperature at 420^{0}C and vice vera.
- Auctioneering Control System is applied for this purpose.

For Pressure Control:

- Control Objective is to maintain the Reactor Pressure and prevent Pressure Buildup.
- Controlled Variable is Pressure of the Reactor.
- Manipulated Variable is the Pressure Relief Valve of the outlet stream of the Reactor.
- Control Action is, when Pressure of the reactor rises, the Valve opens to remove excess pressure buildup in the reactor.
- The Feed-Backward control system is used for this purpose.

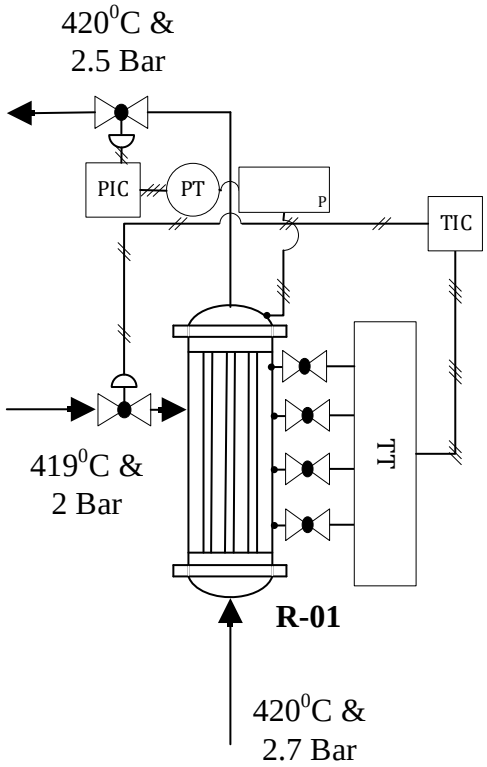


Figure 5.2: Control Loops on Reactor

5.2.6. Process Control Configuration for the Flash Vessel

The Pressure Buildup is prevented by the help of a sensor on the Flash Vessel (Differential Pressure Cell), a controller and hence the Pressure Relief Valve. In this case,

For Pressure Control:

- Control Objective is to maintain the Vessel Pressure and prevent Pressure Buildup.
- Controlled Variable is Pressure of the Vessel.
- Manipulated Variable is the Pressure Relief Valve of the Outlet Gases.
- Control Action is, when the Pressure of the Vessel rises, the outlet gas stream Valve opens to remove excess pressure buildup in the Vessel.
- The feed-Backward Control system is used for this purpose.

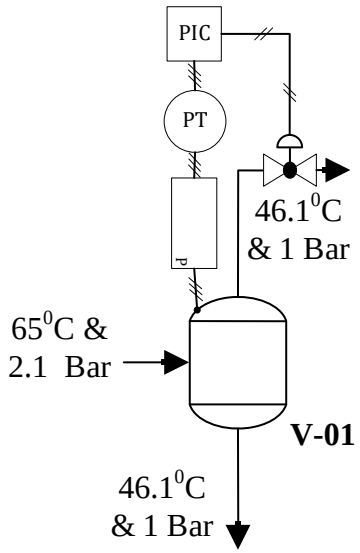


Figure 5.3: Control Loop on Flash Vessel

5.2.7. Process Control Configuration for the Absorber

The Pressure Buildup is prevented by the help of a sensor (Differential Pressure Cell), a controller and hence the Pressure Relief Valve. The level is maintained by manipulating the outlet liquid flowrate (Feed-Forward Control System). The ratio of the inlet flowrates is controlled by manipulating the independent stream flowrate (Ratio-Control System). In this case,

For Temperature Control:

- Control Objective is to maintain the Column Temperature to achieve the required separation, based upon the solubility at that temperature.
- Controlled Variable is Temperature of the Column.

- Manipulated Variable is the Valve of the inlet solvent stream.
- Control Action is, when temperature of the column rises, the flowrate of the solvent increases to maintain the temperature and vice vera.
- Auctioneering Control System is applied for this purpose.

For Pressure Control:

- Control Objective is to maintain the Column Pressure and prevent Pressure Buildup.
- Controlled Variable is Pressure of the Column.
- Manipulated Variable is the Pressure Relief Valve of the Column.
- Control Action is, when Pressure of the Column rises, the outlet gas stream Valve opens to remove excess pressure buildup in the Column.
- The feed-Backward Control system is used for this purpose.

For Level Control:

- Control Objective is to maintain the liquid level on trays in the column.
- Controlled Variable is Level of Liquid in the column
- Manipulated Variable is the Outlet flowrate of the Liquid.
- Control Action is, when level is disturbed, the outlet flowrate is manipulated to maintain the necessary level.
- The feed backward control system is used for this purpose.

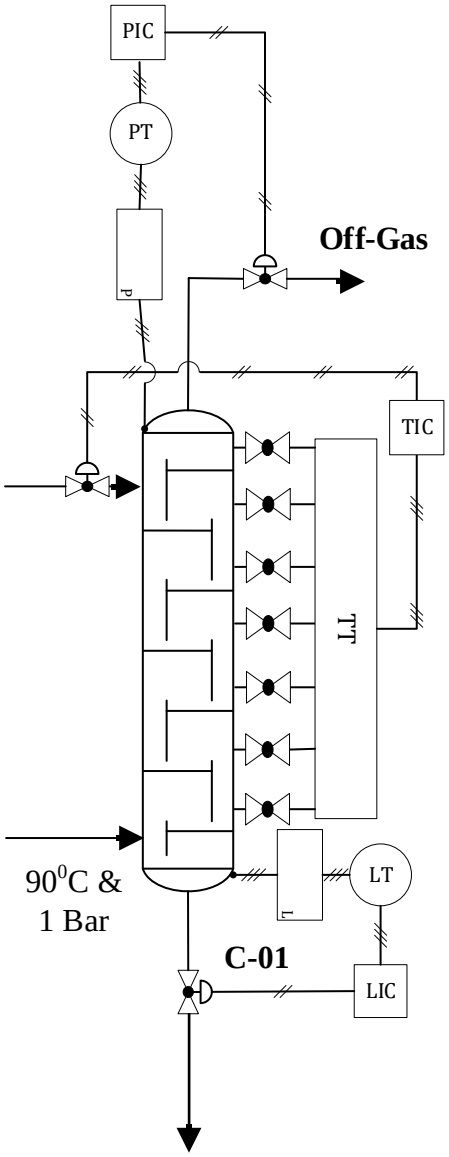


Figure 5.4: Control Loops on Absorber

5.2.8. Process Control Configuration for Distillation Column

The Pressure of the Column is to be maintained and the pressure buildup is prevented with help of Pressure relief valve. The level of the liquid on the trays is maintained with the help of Level sensors and controllers and then manipulating the liquid outlet flowrate. The Temperature of the cooler stream is maintained by manipulating the flowrate of the Cooling water. The flowrate of the reflux is maintained by manipulating the distillate flowrate. The reboiler stream flowrate is maintained by manipulating the

bottom product flowrate. The reboiler temperature is controlled by manipulating the high-pressure stream flowrate. In this case,

For Pressure Control:

- Control Objective is to maintain the Column Pressure and prevent Pressure Buildup.
- Controlled Variable is Pressure of the Column.
- Manipulated Variable is the Pressure Relief Valve of the Column.
- Control Action is, when Pressure of the Column rises, the Valve opens to remove excess pressure buildup in the Column.
- The feed backward control system is used for this purpose.

For Level Control:

- Control Objective is to maintain the liquid level on trays in the column.
- Controlled Variable is Level of Liquid in the column
- Manipulated Variable is the Bottom flowrate of the Liquid.
- Control Action is, when level is disturbed, the Bottom flowrate is manipulated to maintain the necessary level.
- The feed backward control system is used for this purpose.

For Temperature Control of the Cooler Outlet Stream:

- Control Objective is to maintain the Cooler outlet stream temperature.
- Controlled Variable is Temperature of the Cooler outlet stream.
- Manipulated Variable is the flowrate of the Cooling water.
- Control Action is, when temperature of the outlet stream rises, the flowrate of cooling water increases to maintain the temperature of the cooler outlet stream and vice versa.
- The feed backward control system is used for this purpose.

For Reflux Flowrate Control:

- Control Objective is to maintain the reflux flowrate of the column.
- Controlled Variable is Flowrate of the reflux.

- Manipulated Variable is the flowrate of the Distillate.
- Control Action is, when the Reflux flowrate increases, the distillates flowrate increases to maintain the reflux flowrate at the set point and vice versa.
- The feed backward control system is used for this purpose.

For Reboiler outlet stream Flowrate Control:

- Control Objective is to maintain the Reboiler outlet stream flowrate.
- Controlled Variable is Flowrate of the reboiler outlet stream.
- Manipulated Variable is the flowrate of the Bottom Product.
- Control Action is, when the Reboiler outlet flowrate increases, the Bottom Product Flowrate increases to maintain the Reboiler outlet flowrate at its set point and vice versa.
- The feed backward control system is used for this purpose.

For Temperature Control of the Reboiler Outlet Stream:

- Control Objective is to maintain the Reboiler outlet stream temperature.
- Controlled Variable is Temperature of the Reboiler outlet stream.
- Manipulated Variable is the flowrate of the High-pressure steam.
- Control Action is, when temperature of the reboiler outlet stream rises, the flowrate of High-pressure stream decreases to maintain the reboiler outlet stream temperature and vice versa.
- The feed backward control system is used for this purpose.

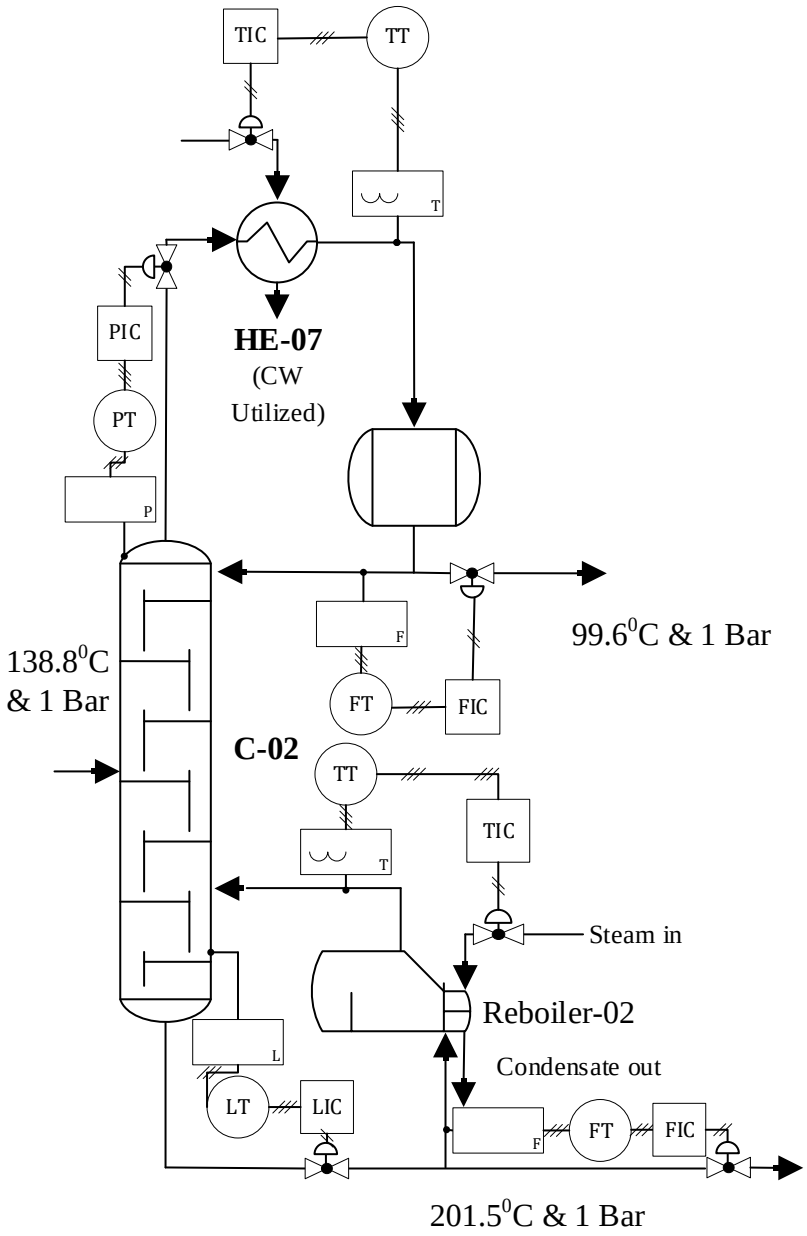


Figure 5.5: Control Loops on Distillation Column

Chapter 6

Material of Construction

Selection of materials the chemical engineer responsible for the selection of material of construction must have thorough understanding of all the basic information about equipment add also have fully information about the material uses. There are some key points given below for material of selection

Preliminary selection

Experience, special literature, availability, safety aspect, manufacture data, preliminary laboratory tests.

Laboratory testing

Reevaluation of apparently suitable materials under process conditions like temperature, pressure, toxic etc.

Interpretation of laboratory results and other data

Effects possible impurities, access pressure, access temperature, ignition, and presence of air in equipment, avoidance of electrolysis and fabrication methods.

Economic comparison

Material and maintenance cost, cost of product degradation, liability of special hazards and probable life

Guideline for selection of material:

For the selection of proper material there are four elements that should be considered highly

- Safety of the Equipment (considering operators and other plant personnel).
- Strength resistance to sudden failure from mechanical or thermal shock, corrosion resistance.
- Chemical compatibility
- Total Cost
- Expected operating life

6.1. Feed Salt Heat Exchanger

For Feed Salt heat exchanger (shell and tube type) , we must use a material i.e., corrosion resistant and have good thermal conductivity. These two factors will improve the efficiency of heat exchanger and will make its life longer and when the material is corrosion resistant, it will not deteriorate early and thus less maintenance cost will be required.

For Heat Exchanger, Stainless Steel, Carbon Steel, and Copper Alloys (i.e., Cu-Ni, and others) are preferable. But in these Stainless Steel is the ideal material for heat exchanger construction. Since,

- Stainless Steel has proven to be a reliable and durable material. It is resistant to corrosion and deposits of residues like limestone etc. are minimized.
- Also, Stainless Steel have 10.5% more chromium than other alloys which makes it incredibly resistant to corrosion and rust than other materials.
- In comparison to copper used for heat exchanger construction, Stainless steel is on average 2.5 times stronger than copper and it performs much better on stress and strain tests. The weaker structure of copper makes its piping unprotected to external forces that may cause cracks and leaks.

Thus, Stainless steel 304/304L, 347 and 410 will be a good fit for shell and tube heat exchanger.

6.2. Reactor

For Reactor (Packed Bed here), Carbon steel and Stainless steel are preferable for construction with respect to the operating conditions (i.e., temperature, pressure involved). But in both, Stainless steel is much preferable due to the following reasons:

- Stainless steel is more corrosion resistant, especially in oxidizing environment due to content of chromium present (usually >12%), as compared to Carbon steel that when exposed to moisture can corrode and rust quickly.
- Although Carbon steel is stronger, more durable, and cheaper than Stainless steel, but the increased corrosion resistance of Stainless steel increases the operating life of Reactor and also lessen the maintenance cost (as in our case oxidizing environment is involved). Thus, in terms of economic analysis,

although Carbon steel has less first cost but more operating cost and lesser expected life.

Thus, Stainless Steel (mainly 304L and 316L grades) will be the good fit for the reactor as they are most effective in hot and corrosive environments.

6.3. Distillation Column, Absorber & Flash Vessel

Mostly for the fabrication of absorber and distillation column stainless steel and carbon steel is used.

- Stainless steel has high chromium content (equal to or greater than 10.5% chromium) which act as a protective layer against corrosion and rust.
- Carbon steel is high in carbon that when exposed to moisture can rust and corrode quickly. but to overcome that We can make a coating of chromium on carbon steel.
- Carbon steel is stronger and more durable than stainless steel.
- Stainless steel can operate better at high pressure and temperature as it has high strength and durability.
- Carbon steel is cheaper than stainless steel.
- It has no effect of toxic materials.

As we have high temperature and pressure is 1 bar, so it is better to use carbon steel for the fabrication of absorber and distillation column.

Table 6.1: Material of Construction for Flash Vessel, Distillation Column and Absorber

Parameters	**Distillation column**	**Absorber**	**Flash vessel**
Inlet Temperature	138.8^0C	90^0C	65^0C
Inlet Pressure	1 bar	1 bar	2.1 bar
Inlet Flow rate	8759.479 kg/hr.	83451.55 kg/hr.	86642.49 kg/hr.
Nature of fluids	Toxic	Toxic	Toxic
Material for Fabrication	Carbon Steel coated with chromium	Carbon Steel coated with chromium	Carbon Steel coated with chromium

Chapter 7

HAZOP Study

HAZOP Study (Hazards and Operability study) is the study of the hazards, their causes, effects, and their control measures to be taken in the process. The following is a detailed HAZOP analysis on each designed equipment in this production plant design is discussed:

7.1. HAZOP Analysis on Reactor

Two study nodes connected to reactor is selected to be studied in this HAZOP analysis. The study nodes are shown in the table and figure below.

Table 7.1: Study Nodes in HAZOP Analysis

No.	**Study node**
1.	Inlet to reactor (K-1) – Stream A
2.	Outlet to reactor (K-1) – Stream B

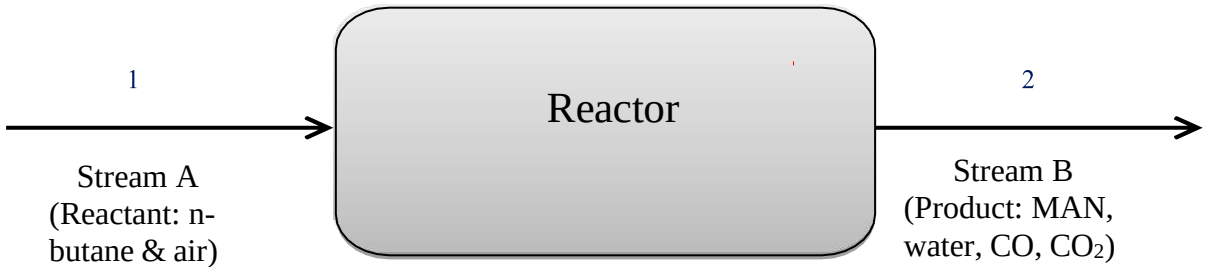


Figure 7.1: Selected Study Nodes connected to Reactor

The given tables illustrate findings from HAZOP analysis on Reactor:

Table 7.2: HAZOP study on Reactor (Stream A)

HAZOP STUDY RECORD SHEET		**PROJECT: MALEIC ANHYDRIDE PLANT** **MAJOR UNIT: REACTOR**		**Date: 22 May, 2022**
System: MAN Production		**Study Note: Inlet to reactor (K-1) -- Stream A**		
Guide Word	**Deviation**	**Possible Consequences**	**Possible Causes**	**Safeguard/ Action Required**
Flow	More	• Upsetting Process • Uncontrolled reaction • Excessive heat	• Excess air • Excess n-butane • Valve fully open	• Install High Flow Alarm • Inspect/repair/change valve
	No	• No reaction • No product	• Pipe rupture • Valve fully closed • Blockage	• Install feed flow meter • Regular inspection and maintenance on equipment
	Less	• Less Pressure • Less favorable product	• Pipe leakage • Valve partially closed	• Install Low Flow Alarm • Regular inspection and maintenance on equipment • Inspect/repair/change valve

Pressure	More	• High Velocity of reactant • Overpressure • High Temperature	• Valve failure • Overheat • Dryer failure	• High pressure alarm • Install safety valve • Install high temperature alarm
	Less	• Less reaction • Less favorable product	• Leakage • Compressor failure	• Insert safety valve • Install low pressure alarm
Ignition		• Fire, Sparks	• Presence of oxidizing substance • Temperature rise	• Insulation of pipe and equipment • Temperature sensor
Temperature	More	• Uncontrolled reaction • Mechanical failure	• Pressure increase	• Install temperature sensor
	Less	• Less pressure	• Low Pressure	• Install sensor

Table 7.3: HAZOP study on Reactor (Stream B)

HAZOP STUDY RECORD SHEET		**PROJECT: MALEIC ANHYDRIDE PLANT** **MAJOR UNIT: REACTOR**		**Date: 2 November, 2021**
System: MAN Production		**Study node: Outlet to reactor (K-1) – Stream B**		
Guide Word	**Deviation**	**Possible Consequences**	**Possible Causes**	**Safeguard/ Action Required**
Flow	More	• Upsetting process • Increase demand for solvent	• Increase in temperature • Uncontrolled reaction	• Install high flow alarm
	No	• No product	• Pipe rupture, valve fully closed, blockage • No reaction	• Regular inspection and maintenance on equipment • Inspect/repair/change valve
	Less	• Less product • Decrease in temperature	• Low reaction rate • Pipe leakage • Valve partially close	• Install low flow alarm • Regular inspection and maintenance on equipment • Inspect/repair/change valve
Pressure	More	• Low absorption rate	• Uncontrolled reaction	• Install safety valve • Install high pressure alarm

	Less	• Less flow rate to the absorber • Agitate absorption process • Take longer time	• Low reaction rate • Pipe leakage	• Install low flow alarm • Regular inspection and maintenance on equipment
Temperature	More	• Increase in utility	• Uncontrolled reaction	• Install temperature sensor • Install auto-control valve for utility
	Less	• Less pressure • Take longer time for absorption process	• Low reaction rate.	• Install temperature sensor

7.2. HAZOP Analysis on Feed Salt Heat exchanger

The given table illustrates findings from HAZOP analysis on Feed Salt Heat Exchanger:

Table 7.4: HAZOP study on Feed Salt Heat Exchanger

Study Node	Parameter	Guide Word	Deviation	Possible cause	Consequences	Safeguards	recommendations
Heat exchanger	flow	NO	No Flow	Pump faulty	No heat exchange occurs	Fail open valve	Create emergency response
				Pipe blockage	Economic loss	Flow indicator	Check the condition of the equipment and the pipeline
		Less	Less Flow	Feed control valve partially closed	Less yield of desired product	Flow control valve	Regular inspection of the flow control valve
				Leakage in inlet pipeline	Production loss	Low flow alarm system	Install low flow alarm at inlet of the steam flow
		More	More Flow	Feed control valve fail open	Change in product composition	High flow alarm system	Switch valve to control manually

7.3. HAZOP Analysis on Distillation Column:

The following table illustrates the HAZOP study on the distillation column:

Table 7.5: HAZOP Study on the Distillation Column

Guide Word	Deviation	Possible Causes	Consequences	Actions Required
None	No Flow	No flow in stream coming to the column	Quality and amount of product will suffer.	Install low level alarm
		Line fracture	Accidental discharge to the environment (aromatics are flammable material).	Plant shut down.
More of	Flow	LCV fails open in error	Overfills Incomplete separation of non-aromatics from solvent + aromatics	Install high level alarm and check sizing.
	Pressure	Isolation valve close in error	- Full pump delivery	- Isolation valve close in error
	Temperature	Higher pressure in transfer line	Degradation of the solvent quality will become significant ms.	High Temperature Alarm (HTA).
Less of	Flow	Leaking flange of valve	Material discharge to the environment	Isolate that part of the plant
Others	Maintenance	Equipment Failure	Line cannot be completely drained or purged	- Install alarms.

Chapter 8

Cost Estimation

8.1. Cost Estimation Methods

This relation is used to estimate the cost. To incorporate the recent increases in prices a factor of 1.2 or 1.5 can be used according to inflation of prices in Pakistan

$$\mathbf{C_e} = \boldsymbol{a} + \boldsymbol{b}(\boldsymbol{S})^{\boldsymbol{n}}$$ (Gavin Towler n.d.)

Where,

Ce = purchased equipment cost on a U.S. Gulf Coast basis, January 2006 (CE index ¼ 478.6, NF refinery inflation index ¼ 1961.6);

a, b = cost constants in Table 6.6;

S = size parameter, units given in Table 6.6;

n = exponent for that type of equipment.

8.1.1. Cost Estimation for Pump

Volumetric flow rate = 0.001323 L/s,

Stainless steel 304 Cost for pump =

$Ce = a + b\ S^{n}$

Centrifugal pump single stage: $Ce = a + b(S)^{n}$

$Ce = 61000 + 650(0.001323)^{0.9}$

Ce = $61001.67

Material factor = 1.3

Ce = 61001.67× 1.3 = $ 79,302.17

8.1.2. Cost Estimation for Compressor

Flow rate = 83911.6225 kg/hr

Density of air = 1.184 kg/m^3

Volumetric flow rate = 70871.3 m^3/hr

By using HYSYS, we get the purchase cost of Air Compressor for the year 2013 as;

Purchase Cost = $39743

This is the cost in year 2013.

The average increase in the cost is about 2.5% per year.

Using this value we predict the heat exchanger purchased cost in year 2022

In year 2022;

Purchased cost = 39743 × $(1.025)^9$

Purchased cost = $49633.56

8.1.3. Cost estimation for Distillation column

Material of construction is carbon steel

Pressure in Column= 1 bar = 100000 N/m^2

Diameter of Column = 1.095m

Length of Column= 9 m

Maximum Allowable Pressure = 120000 N/m^2

Column Mass = ρ*Dc*Lc*tw*π

Dc = vessel diameter,

L_c = vessel length,

T_w = wall thickness,

ρ = metal density, kg/m^3.

We assume the welding efficiency is .85 because of spot radiographed

Efficiency = E = 0.85

S =12.9 KSI @300-500 F (from ASME BPV Code Sec. VIII D.1, Taken From Sec. II Part D)

= $88.9x10^7$ N/m^2

We need to keep in mind that we are using carbon steel whose

We need to find the minimum thickness of the sheet so we can calculate Mass

$$T_w = \frac{P*D}{2SE-(P)} = \frac{100000*1.095}{(2*8.89*10^7*0.85)-(1.2*100000)} = 0.00072 \text{ m} = 0.72 \text{ mm}$$

Mass of Column = π * 0.00072 * 1.095 * 9 * 7840 = 174.7652 kg

Purchased Cost of Column= a + b $(S)^n$

Purchased Cost of Column = -400 + 230 $(174.7652)^{0.60}$ = $4695.664

Material Cost Factor of carbon Steel = 1.0

Collective Cost of Column including Purchased Cost and Material Cost = 1.0* 4695.664= $4695.664

Installation factor of Column= 4 (Ray sinnot 6.3)
(Sinnott, Galvin Towler and Ray 2002)

Total Cost of Column = 4* 4695.664= 18782.66

No. of Trays = 15

Purchased Cost of one tray = a + b $(S)^n$

As we are using Sieve trays. So,

as we have the diameter of column=1.095 m

Purchased Cost of one tray = 100 + 120 $(1.095)^2$ = $243.883

Purchased Cost of all trays = 15 * 197.2 = $3658.245

Material Cost Factor of Carbon Steel = 1.0

Collective Cost of trays including Purchased Cost and Material Cost = 1 * 2958 = $2958

Installation factor of trays= 2.5 (Miscellaneous 6.3)

Total Cost of trays = 2.5 * 3658.245 = $9145.613

Total Cost of Binary Distillation Column =9145.613+18782.66 = $27928.3

But this cost is according to 2006 but we need it according to present price in order to do so we will use the dollar price index

Installed cost $= \frac{27928.3*610}{478.6}$

= $35596.04

8.1.4. Cost Estimation for Feed Salt Heat Exchanger

As we know the area of shell and tube heat exchanger calculated was 51.4 m^2

Now estimating the cost:

$C_e = a + b(S)^n$

$C_e = 10000 + 88(51.4)^1$

= $14523.2

Since the material of construction is stainless steel so:

Material Cost Factor of Stainless Steel = 1.5

Collective Cost of heat exchanger including Purchased Cost and Material Cost = 1.5* 14523.2= $21784.8

Installation factor of Column= 3.5 (Ray Sinnot table 6.3) (Sinnott, Galvin Towler and Ray 2002)

Total Cost of Column = 3.5* 21784.8= $76246.8

But this cost is according to 2006 but we need it according to present price to do so we will use the dollar price index

Installed cost $= \frac{76246.8*610}{478.6}$

= $97180.42

8.1.5. Cost estimation for Flash column

As we know the area of Flash column calculated was 5.19 m^2

Pressure = 2.1 bar = 210000 N/m^2

Material of construction is carbon steel

Diameter of Column = 2.5 m

Length of Column=7.7 m

E = 1

S =12.9 KSI @100-500 F (from ASME BPV Code Sec. VIII D.1, Taken From Sec. II Part D)

= $88.9x10^7$ N/m^2

Column Mass = ρ*Dc*Lc*Tw*π

Dc = vessel diameter,

L_c = vessel length,

T_w = wall thickness,

ρ = metal density, kg/m^3.

We assume the welding efficiency is 1 because of full radiographed welding

Efficiency = E = 1

We need to find the minimum thickness of the sheet so we can calculate Mass

$T_w = \frac{P*D}{2SE-(P)} = \frac{210000*2.57}{(2*8.89*10^7*1)-(1.2*210000)}$ = 0.0014464 m = 1. 45 mm

Column Mass = π*2.57*7.7*0.00145*7840

= 706.73 kg

Now estimating the cost;

$C_e = a + b(S)^n$

C_e = -400 + 230(706.73)$^{.6}$

= $11383.65

Since the material of construction is Carbon Steel so;

Material Cost Factor of carbon Steel = 1

Collective Cost of Column including Purchased Cost and Material Cost = 1* 11383.65

= $11383.65

Installation factor of Column= 2.5 (Gavin Towler n.d.)

Total Cost of Column = 2.5* 11383.65= 28459.13

But this cost is according to 2006 but we need it according to present price in order to do so we will use the dollar price index

Installed cost $= \frac{28459.13*610}{478.6}$

= $36272.60

8.1.6. Cost estimation for Absorber

As we know the area of Absorption column calculated was 1.75584 m^2

Pressure = 1 bar = 100000 N/m^2

No. of Trays = 15

We are using sieve trays.

Material of construction is carbon steel

Diameter of Column = 1.4956 m

Length of Column= 7.8 m

S =12.9 KSI @100-500 F (from ASME BPV Code Sec. VIII D.1, Taken From Sec. II Part D)

= $88.9x10^7$ N/m^2

Column Mass = ρ*Dc*Lc*Tw*π

Dc = vessel diameter,

L_c = vessel length,

T_w = wall thickness,

ρ = metal density, kg/m^3.

We assume the welding efficiency is 1 because of full radiographed welding

Efficiency = E = 1

We need to find the minimum thickness of the sheet so we can calculate Mass

$T_w = \frac{P*D}{2SE-(P)} = \frac{100000*1.4956}{(2*8.89*(10)^7*1)-(1.2*100000)}$ = 0.000842 m = .8 mm

Column Mass = π*1.4956*7.8*0.000842*7840

= 241.9291 kg

Now estimating the cost;

$C_e = a + b(S)^n$

C_e = -400 + $230(241.9291)^{.6}$

= $5793.565

Installation factor of Column= 2.5 (Ray sinnot 6.3)

Total Cost of Column = 2.5* 5793.565= 14483.9

As we are using Sieve trays. So,

as we have the diameter of column=1.4956 m

Purchased Cost of one tray = 100 + 120 $(1.4956)^2$ = $368.4183

Purchased Cost of all trays = 13 * 368.4183= $4789.438

Installation factor of Column= 2.5 (Ray sinnot 6.3)

Total Cost of Column = 2.5* 4789.438= $11973.59

Now Total cost of Absorber with the tray cost = 11973.59 + 14483.9 = 26457.5

But this cost is according to 2006 but we need it according to present price in order to do so we will use the dollar price index

Installed cost $= \frac{26457.5*610}{478.6}$

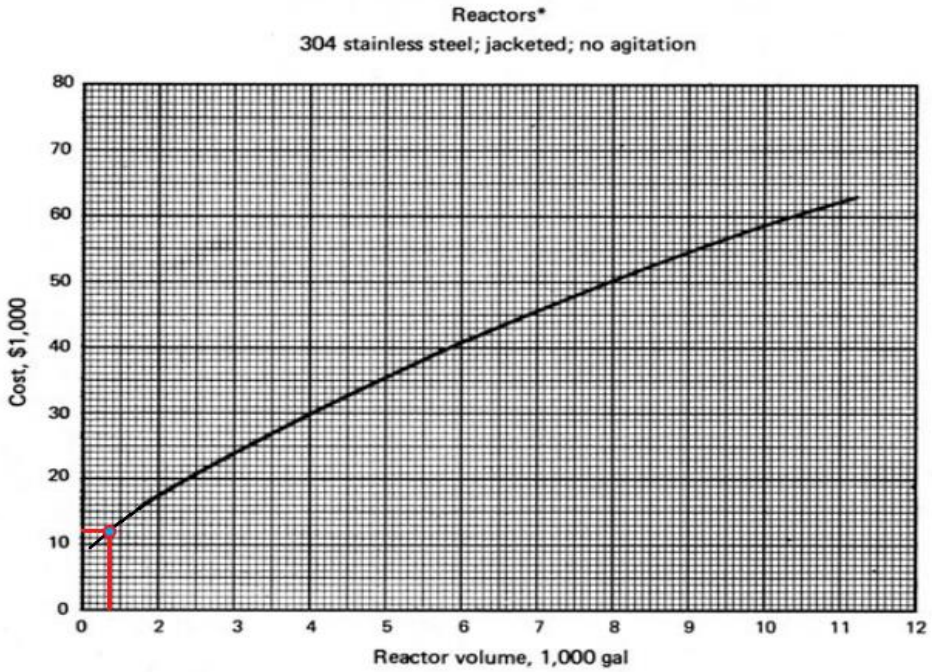
= $33721.43

8.1.7. Cost Estimation for Reactor

This figure is used to estimate the cost of a non-agitation reactor.

Formula of calculation is as follows;

installed cost = purchase price x installation factor



Module factors:		Installation factor:	Material factors:	
Stainless	1.8	1.40 – 2.10; avg. 1.70	316 stainless	1.2
Glass lined	2.1		Glass lined	.8
Mild steel	2.3		Lead lined	.7
			Mild steel	.6

Figure 8.1: Graph of Cost Vs Capacity

Purchased price = $12000

Installation factor = 1.9

Installed cost = $12000 * 1.9

= $22800

Material Cost Factor of Carbon Steel = 1

Installed cost with Material cost factor = 1.5*22800

= $22800

But this cost is according to 1989 but we need it according to present price to do so we will use the dollar price index

Installed cost $= \frac{\$22800*610}{93.93}$

= $148067.7

Equipment Name	No of Equipment	Cost of Per Equipment ($)	Total Cost of specific equipment
Pump	**2**	79,302.17	158604.3
Compressor	**1**	49633.56	49633.56
Heat Exchangers	**9**	97180.42	874623.8
Reactor	**1**	148067.7	148067.7
Absorber	**1**	33721.43	33721.43
Distillation	**1**	35596.04	35596.04
Flash Vessel	**1**	36272.60	36272.6
Total	**16**	479,773.92	1,336,519

Here we will estimate the cost using Hand's method and the factors that are specified for

Hand's method.

ISBL = ∑Ce = $1,336,519

ISBL = $ 1.3× 10^6

ISBL = $ 1.3 MM ± 30% (within the accuracy of the method)

OSBL investment

Offsite investments often involve interactions with utility companies such as electricity or water suppliers. Offsite costs are typically estimated as a proportion of ISBL costs in the early stages of design. For typical chemicals projects, offsite costs are usually between 20% and 50% of ISBL cost, and 40% is usually used as an initial estimate.

OSBL = 40 % (ISBL)

OSBL = 40 % ($1,336,519) = $5, 34,607.78

Engineering Cost

Engineering costs are best estimated individually based on project scope, as they are not directly proportional to project size. A rule of thumb for engineering costs is 30% of ISBL for smaller projects and 10% of ISBL for larger projects.

We consider 80 tons per day a part of a large project so we are taking it as 10 percent of the ISBL

E. C = 10 % (ISBL)

E. C. = 10 % ($1, 336,519) = $ 1, 33,651.95

Contingency Cost

A contingency charge can be thought of as an additional fee charged by the Engineering, Procurement, and Construction (EPC) Company to address the risk that the project will run over budget. A minimum contingency charge of 10% of ISBL should be used on all projects

C. C. = 10 % (ISBL)

C. C. = 10 % ($1,336,519)

= $1, 33,651.95

Therefore, fixed capital cost will be:

F. C. C = ISBL + OSBL + E. C. +C. C.

F. C. C= 1,336,519 + 5, 34,607.78 + 1, 33,651.95+ 1, 33,651.95

F. C. C = $ 2,138,431

Working capital Cost (W.C.C)

The capital that is tied up in maintaining inventories of feeds, products, and spare parts, together with cash on hand and the difference between money owed by costumers (accounts receivable) and money owed to suppliers (accounts payable), is termed the working capital of the plant. Working capital is required if the plant is in operation, but it is recovered if the plant is shut down. Working capital should be 10-20% of fixed capital cost used for all projects.

W. C. C = 10 % (F. C. C)

W. C. C = 10 % ($21, 38,431)

W. C. C = $ 2, 13,843

Therefore, total capital cost will be:

T. C. C = F. C. C + W. C. C

T. C. C = 2,138,431+ 2, 13,843

T. C. C = $ 2,352,274

This Total Cost is according to cost index 2022. If we want to convert it into PKR so according to 15, June, 2022 the cost comes out to be

T. C. C = $\frac{\$\ 2{,}352{,}274 * \text{PKR}\ \$\ 206.27}{\$1}$

= 485,203,558 PKR

= 485 Million Pakistani Rupees

Chapter 9

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